MATH II

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Subject materials:

- ⊕ Folder with squared sheets.
- ⊕ Pencil, eraser, protactor, and ruler.
- ⊕ Calculator
- ⊕ Printed booklet.

EVERYONE SHOULD HAVE THE MATERIALS FOR ALL CLASSES.

TESTS

Date	Topics

Course Objective

- Get the students prepared to take the S.A.T.
- Understand the significance and meaningful concepts, as well as connections among numeric fields or geometric concepts.
- Be able to transfer to daily life the contents given (including topics, problems and issues related to cross-curricular projects: Sustainable Development Goals and E.S.I.)
- Integrate the knowledge acquired in previous years with the new information.
- Promote respect to diversity and equality (E.S.I)

Unit Titles/Topics/Brief Description

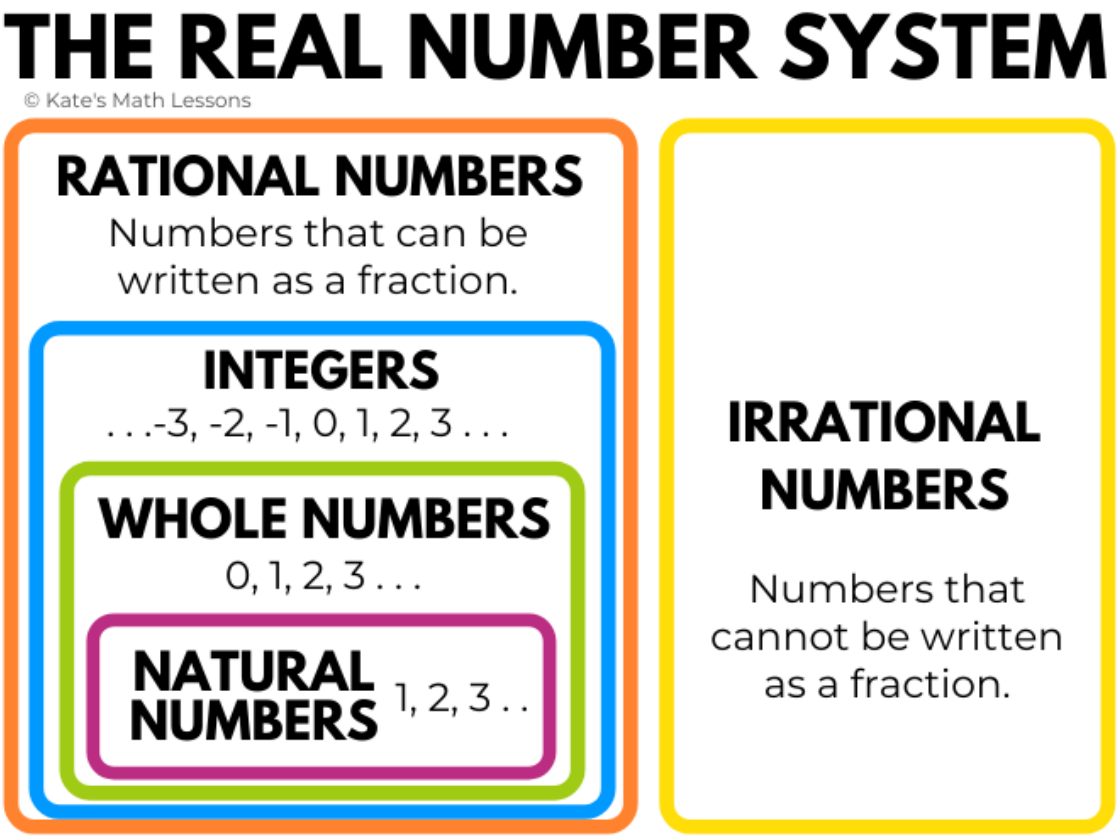
Unit 0	**DIAGNOSTIC UNIT**
• Operations with whole numbers. • Properties of operations with whole numbers. • LCM, GCF	
Unit 1	**NUMBERS AND OPERATIONS**
• Integer numbers. • Operations with integers. • Number lines with integer numbers. • Roots and exponents. • PEMDAS • Fractions. Operations with fractions. • Arithmetic word problems	
Unit 2	**ALGEBRA AND FUNCTIONS**
• Introduction to algebra • Two-dimensional coordinate system. Locations. • Linear equations.	
Unit 3	**GEOMETRY AND MEASUREMENT**
• Angles in a plane. Definition, basic notation, and main properties. • Special angles: adjacent, complementary, supplementary, and vertical angles. Angles with parallel lines. • Polygons. Area and perimeter.	
Unit 4	**DATA ANALYSIS, PROBABILITY AND STADISTICS**

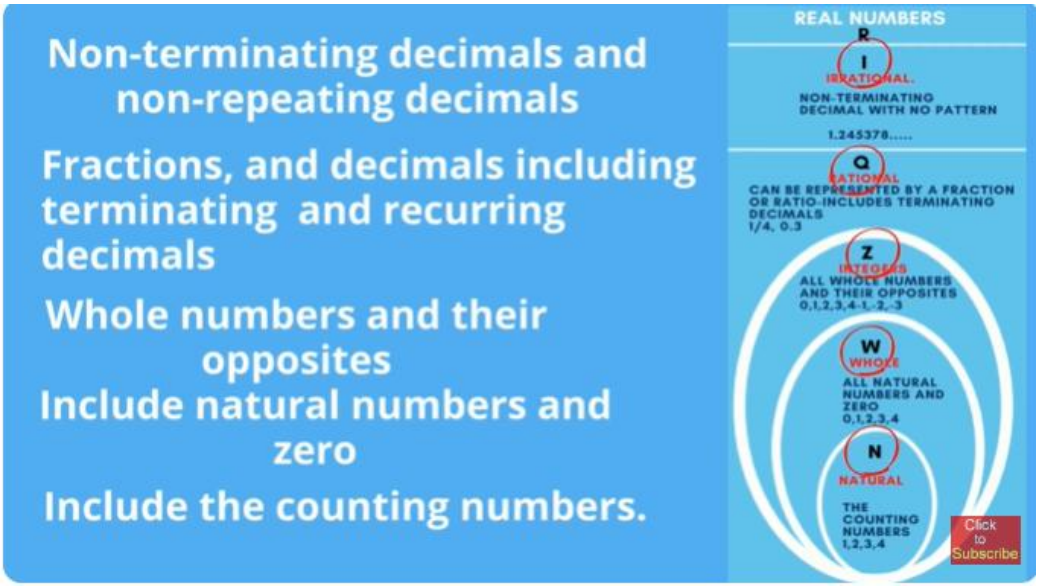
- Data interpretation.
- stadistics. Arithmetic Mean, mode, and media. Frequency.
- Interpretation of information presented in stadistical graphs.
- Probablity

UNIT 0: DIAGNOSTIC UNIT

NATURAL AND WHOLE NUMBERS

Symbol of the Set of Natural Numbers
$\mathbb{N}$
Elements of the Natural Numbers
{ 1, 2, 3, 4, 5, 6, … }





What are the Properties of Whole Numbers?

Whole numbers are countable numbers that include zero and are generally used for calculation purposes. The set of whole numbers in Mathematics starts from 0, that is {0, 1, 2, 3, ...}. The set of whole numbers is denoted by the symbol **W**.

The properties of whole numbers are as follows:

- Closure Property
- Associative Property
- Commutative Property
- Distributive Property
- Identity Property

PROPERTIES OF OPERATIONS WITH WHOLE NUMBERS

- **Addition**

Closure	Associative	Commutative	Identity
The sum of **two natural numbers** is also a **natural number**. **a + b**	The way in which the addends are grouped does not change the result. **(a + b) + c = a + (b + c)** (2+3) + 5 = 2 + (3+5) 5 + 5 = 2 + 8 10 = 10	The order of the addends does not change the addition. **a + b = b + a** 2 + 5 = 5 + 2 7 = 7	The sum of any natural number and zero is the natural number itself. **a + 0 = a** 6+0=0+6=6 Zero (0) is the additive identity for natural numbers.

- **Subtraction**

No closure	No commutative	Identity
The result of **subtracting two natural numbers** is not always another **natural number**. $2 - 5 \notin N$	$5 - 2 \neq 2 - 5$	The subtraction of any natural number and zero is the natural number itself. **a + 0 = a** 6+0=0+6=6 Zero (0) is the additive identity for natural numbers.

- **Multiplication**

Closure	The result of multiplying **two natural numbers** is another **natural number**. **a · b** $\in$ N
Associative	The way in which the factors are grouped does not change the result. **(a · b) · c = a · (b · c)** (2 · 3) · 5 = 2 · (3 · 5) 6 · 5 = 2 · 15 30 = 30
Commutative	The order of the factors does not change the product. **a · b = b · a** 2 · 5 = 5 · 2 10 = 10

Multiplicative Identity	The **1** is the neutral element of the multiplication because any number multiplied by it gives the same number. **a · 1 = a** 3 · 1 = 3
Distributive	The multiplication of a natural number and a sum is equal to the sum of the **multiplication** of the **natural number** for each of the **addends**. **a · (b + c) = a · b + a · c** 2 · (3 + 5) = 2 · 3 + 2 · 5 2 · 8 = 6 + 10 16 = 16 **Removing a common factor:** It is the reverse of the distributive property. **a · b + a · c = a · (b + c)** 2 · 3 + 2 · 5 = 2 · (3 + 5) 6 + 10 = 2 · 8 16 = 16

- **Division**

No closure	**No commutative**	**Identity**
The result of dividing **two natural numbers** is not always another **natural number**. 2: 6 ∉N	**a: b ≠ b: a** 6:2 ≠ 2:6	If a number is divided by 1, the result is the same number. **a : 1 = a** **8 : 1 = 8**

Properties of operation on whole numbers

	Addition	Subtraction	Multiplication	Division
Closure Property	✓	✓	✓	✗
Commutative Property	✓	✗	✓	✗
Associative Property	✓	✗	✓	✗

1. **Choose the correct option.**

A. 7 + 4 = 4 + 7, Identify the property

a. commutative b. closure c. property of zero d. Associative

B. (a + b) + c = a + (b + c), identify the property.

a. commutative b. closure c. property of zero d. Associative

C. The smallest natural number is:

a. 0 b. 1 c. 2 d. 10

D. Which of the following natural number does not have a predecessor?

a. 0 b. 1 c. 2 d. 999999999

E. In which of the following operation, all three properties i.e closure, commutative and associative will not work?

a. Addition b. Multiplication and division

b. Division only d. Division and subtraction

F. Identify the property: a x 0 = 0 x a = 0

a. closure b. property of zero

c. multiplicative identity d. Associative

G. Given two whole numbers a and b, which of the following may not always be whole numbers?

a. a + b b. a - b c. a x b d. 2a + b

H. Which whole number x does not satisfy the relation x ÷ x = 1

a. 0 b. 2 c. 25 d. 50



- Zero is the smallest whole number
- 1 is the smallest natural number
- All the natural numbers are whole numbers, but all whole numbers are not natural numbers

2. Identify the property used.

a. 56 + 12 = 12 + 56 _____

b. 19 + 0 = 19 _____

c. (45 + 12) + 8 = 45 + (12 + 8) _____

d. 10 x 1 = 1 x 10 = 10 _____

3. Find the successor and predecessor of:

a. 5698 b. 101010 c. 99999

4. Fill in the blanks.

a. 504 + ___ = 112 + 504
b. 101 + 0 = ___ + 101
c. 90 + 43 + 10 + 7 = 100 + ____
d. 112 + (56 + 12) = (___ + 56) + 12

5. Which of the following statement is false? cross it and justify your answer.

a. 16 - 2 = 2 - 16 _____
b. 88 + 4 = 4 + 88 _____
c. 100 - 0 = 0 - 100 _____
d. (19 - 4) - 1 = 19 - (4 - 1) _____
e. 99 x 100 = 100 x 99 _____
f. 20 ÷ 5 = 5 ÷ 20 _____

6. Solve the following using the properties only.

a. 465 + 12 + 45 b. 796 + 245 + 204
c. 25 x 896 x 4 d. 4 x 16 x 25

7. Show that:

a. (865 + 452) + 60 = 865 + (452 + 60)
b. (15 x 20) x 2 = 15 x (20 x 2)
c. 99 + (101 + 102) = (99 + 101) + 102



- Distributive property of addition over multiplication
- a x (b + c) = a x b + a x c

e.g. 9 x (6 + 2) = 9 x 6 + 9 x 2
= 54 + 18 = 72

- Distributive property also work for subtraction
- a x (b - c) = a x b - a x c

- solving using distributive property

12 x 99 = 12 (100 -1) {99 can be written as 100 -1}

12 x 100 - 12 x 1 = 1200 - 12 = 1,188

7 x 102 = 7 x (100 + 2) {102 can be written as 100 + 2}

8 . Fill in the blanks.

a. 27 x 6 + 27 x 9 = 27 (6 + _ _)
b. 52 x 4 - 52 x 3 = _ _ _ (4 - 3)
c. 22 x 12 + 22 x 14 = 22 x _ _ _ _ _ _
d. 5 x 107 = 5 x(100 + _ _)
e. 12 x 98 = 12 x (_ _ _ - 2)

9. Solve uing distributive property only.

a. 5 x 107 b. 12 x 98 c. 18 x 102 d. 30 x 99
e. 30 x 5 + 40 x 5 f. 15 x 12 - 5 x 12 g. 7 x 6 - 7 x 4

GCF and LCM:
CRACK THE CODE

Find the greatest common factor (GCF) or the least common multiple (LCM) of the given sets of numbers. Then, look for any places where that GCF or LCM appears in the code at the bottom of the page, and write the corresponding letter on the line. Use the code to reveal the rest of the joke! **Note:** Not every letter will be used in the code at the bottom.

GCF

B	60 75 GCF: _____	**F**	36 92 GCF: _____
R	28 70 GCF: _____	**O**	48 84 GCF: _____
K	16 50 98 GCF: _____	**H**	32 40 88 GCF: _____

LCM

T	12 15 LCM: _____	**A**	10 18 LCM: _____
N	6 14 LCM: _____	**E**	11 4 LCM: _____
I	6 9 12 LCM: _____	**W**	5 9 15 LCM: _____
P	10 5 3 LCM: _____	**S**	10 20 25 LCM: _____

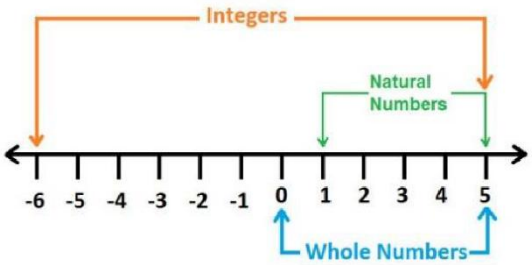
Where do polar bears store their money?

___	___	___	___	___	___	___	___	___	___	___	___	___	___
36	42	60	8	44	100	42	12	45		15	90	42	2

UNIT 1: NUMBERS AND OPERATIONS

INTEGER NUMBERS: number that can be written **without** a fractional component. It includes whole numbers.

Symbol: Z



Number Line representing Integers, Natural & Whole Numbers

Numbers Can be Positive or Negative

This is the Number Line:



Negative Numbers (−) Positive Numbers (+)

−	+
"−" is the negative sign.	"+" is the positive sign

No Sign Means Positive

If a number has **no sign** it usually means that it is a **positive** number.

Example: **5** is really **+5**

integers

Negative integers

numbers less than zero, excluding fractions.

Positive integers

numbers greater than zero, excluding fractions.



No fractions or decimal fractions.

operations on positive and negative integers

Addition

Positive + Positive = Positive
Negative + Negative = Negative
* Positive + Negative or Negative + Positive
  - subtract the smaller number from the larger number, then use the sign of the larger number in the answer

5 + 3 = 8
(- 5) + (- 3) = - 8
(- 5) + 3 = - 2
3 + (- 5) = - 2
(- 3) + 5 = 2
5 + (- 3) = 2

Subtraction

Negative - Positive = Negative
Positive - Negative = Positive
* Negative - Negative = Negative + Positive
  - treat as Negative + Positive
  - subtract the smaller number from the larger number, then use the sign of the larger number in the answer

(- 5) - 3 = (- 5) + (- 3) = - 8
5 - (- 3) = 5 + 3 = 8
(- 5) - (- 3) = (- 5) + 3 = - 2
(- 3) - (- 5) = (- 3) + 5 = 2

Multiplication

Positive x Positive = Positive
Negative x Negative = Positive
Negative x Positive = Negative
Positive x Negative = Negative
- change double negatives to a positive

5 x 3 = 15
(- 3) x (- 5) = 15
(- 3) x 5 = - 15
3 x (- 5) = - 15

Division

Positive ÷ Positive = Positive
Negative ÷ Negative = Positive
Negative ÷ Positive = Negative
Positive ÷ Negative = Negative
- change double negatives to a positive

15 ÷ 3 = 5
(- 15) ÷ (- 3) = 5
(- 15) ÷ 3 = - 5
15 ÷ (- 3) = - 5

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What Did We Find?

Adding a positive number is simple addition ...



Adding a Positive is **Addition**

Positive and Negative Together ...



Subtracting a Positive
or
Adding a Negative
is
Subtraction

Subtracting a negative ...



Subtracting a Negative is the same as **Adding**

The Rules:

It can all be put into **two rules**:

	Rule	Example
+(+) −(−)	Two like signs become a **positive sign**	+ 3+(+2) = 3 + 2 = 5 6−(−3) = 6 + 3 = 9
+(−) −(+)	Two unlike signs become a **negative sign**	− 7+(−2) = 7 − 2 = 5 8−(+2) = 8 − 2 = 6

They are "like signs" when they are like each other (in other words: the same).

So, all you have to remember is:

Two **like** signs become a **positive sign**

Two **unlike** signs become a **negative sign**

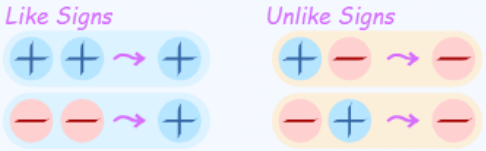
When We Multiply:

		Example
$+ \times +$	two positives make a positive:	$3 \times 2 = 6$
$- \times -$	two negatives make a positive:	$(-3) \times (-2) = 6$
$- \times +$	a negative and a positive make a negative:	$(-3) \times 2 = -6$
$+ \times -$	a positive and a negative make a negative:	$3 \times (-2) = -6$

Yes indeed, two negatives make a positive, and we will explain **why**, with examples!

Two Signs: The Rules

"Two like signs make a positive sign,
two unlike signs make a negative sign"



Dividing Integers

Below is a quick summary for the **rules of dividing integers**.

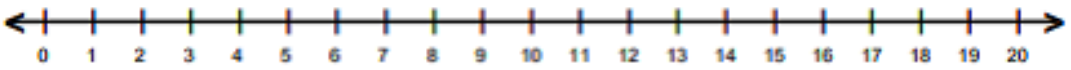
RULES FOR DIVIDING INTEGERS

Signs of Integers	Operation to Use	Answer Sign	Quick Example
⊕ ÷ ⊕	Divide	Positive (+)	$24 \div 6 = 4$
⊖ ÷ ⊖	Divide	Positive (+)	$(-15) \div (-3) = 5$
⊕ ÷ ⊖	Divide	Negative (-)	$42 \div (-7) = -6$
⊖ ÷ ⊕	Divide	Negative (-)	$(-10) \div (5) = -2$

NUMBER LINES AND SCALES

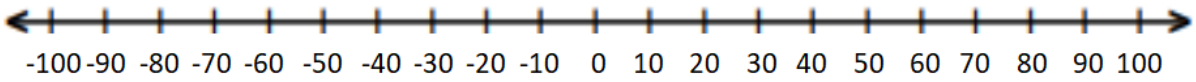
Lines are infinite.

We can draw part of a line and set a scale in order to put numbers in the correct order and make it easier to understand their value.



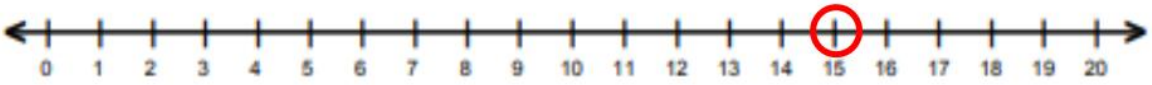
The numbers are often shown as specially-marked points evenly spaced on the line. Although this image only shows the numbers from 0 to 20, the line includes all real numbers, continuing forever in each direction, and also numbers not marked that are between the integers.

The same happens when we work with negative numbers:



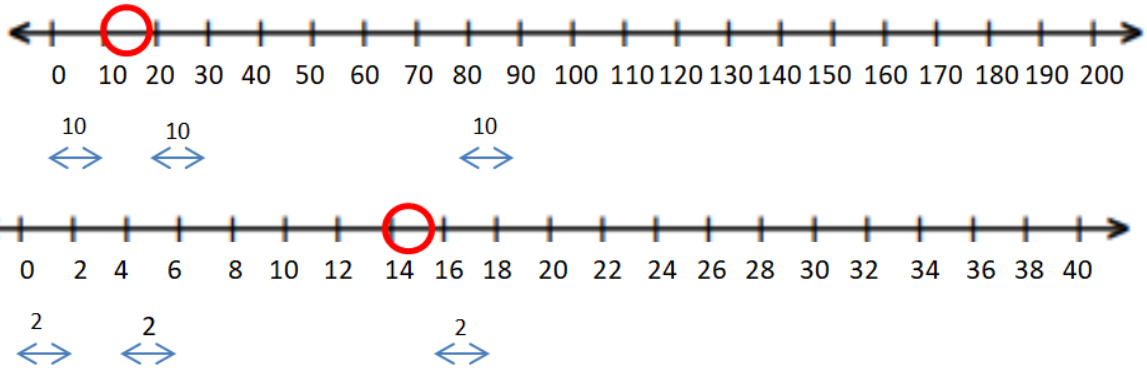
Most of the times, the number we are looking for will not appear on the line, but we can find it by the position it takes and the SCALE that the number line has.

For example: we need to find number 15:

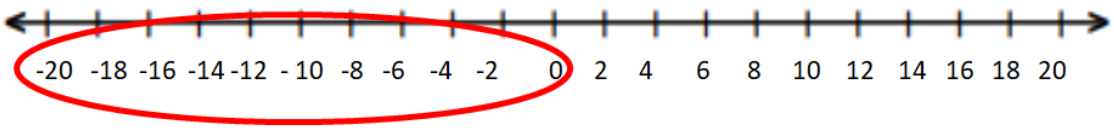


If the scale (value of the distance between tick-marks) changes, then number 15 will be in the middle between 10 and 20 or somewhere between 10 and 12... or somewhere else, depending on the scale we are using.

We can see that the size of the line is the same in both cases, but the value of the distance between tick-marks changes,



When we work with negative numbers, they get place on zero´s left:



I. Write the missing numbers.



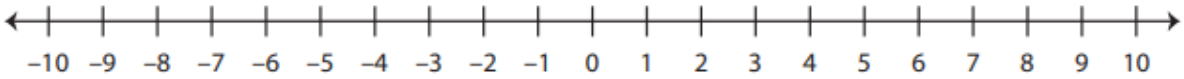
0

2. Arrange the numbers from least to greatest.

5, -2, 0, 8, -3	4, -5, 0, -7, 2
___ , ___ , ___ , ___ , ___	___ , ___ , ___ , ___ , ___

7, -8, -3, 4, -1	-9, 0, -5, 2, 6
___ , ___ , ___ , ___ , ___	___ , ___ , ___ , ___ , ___

3. Answer the questions using the number line below.



1) 2 units to the left of 3 is _____

2) 6 units to the right of –1 is _____

3) 4 units to the left of –4 is _____

4) 3 units to the right of 7 is _____

5) 1 unit to the left of 10 is _____

6) 5 units to the right of –6 is _____

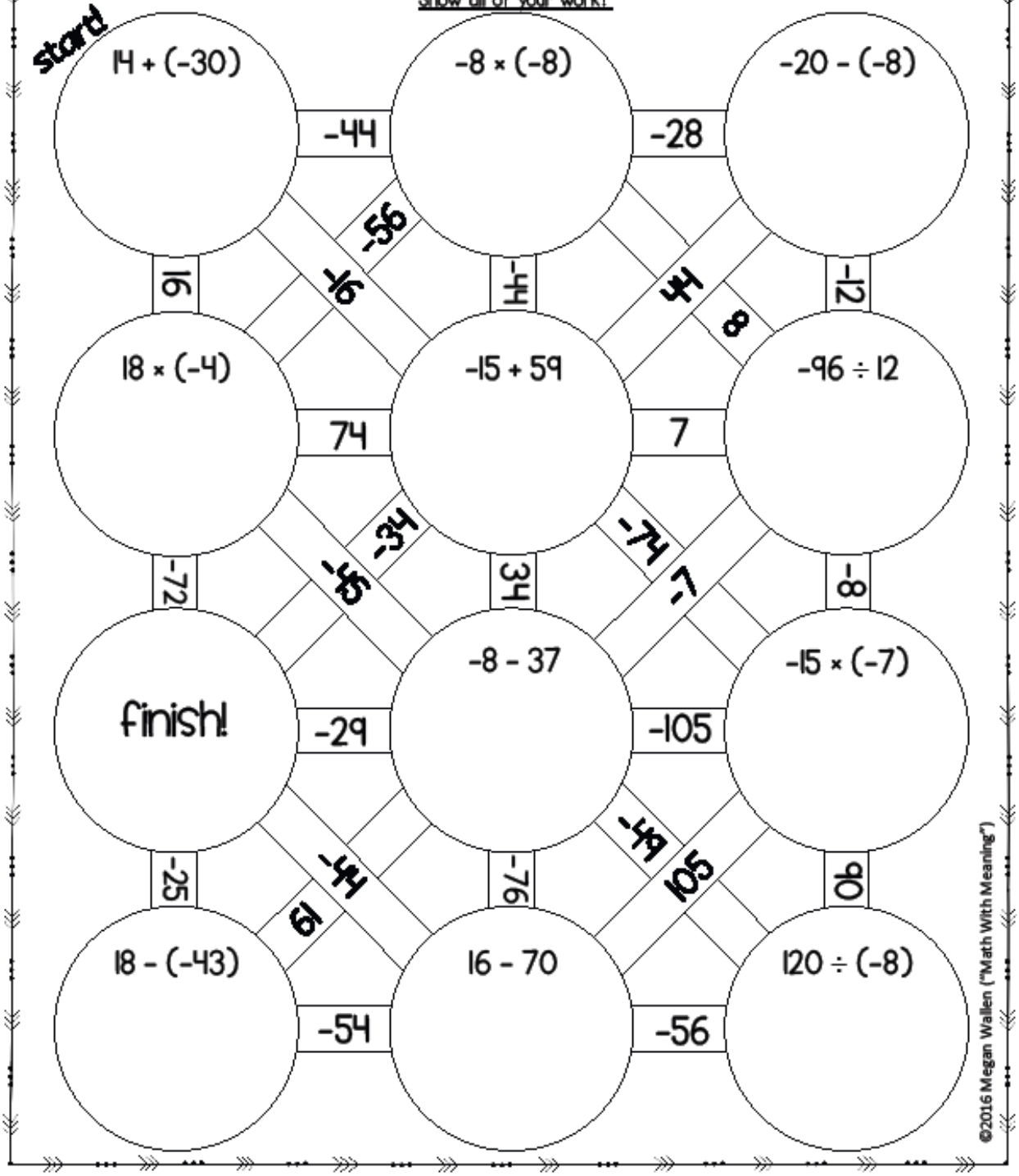
7) 8 units to the left of 5 is _____

Name_____

Integer Operations Maze

Start in the circle marked "Start." Solve the problem, then follow the path with the correct answer to the next problem you will solve. Continue until you reach "Finish." You might not have to solve all problems on the page.

Show all of your work!



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1 Copy the numbers and fill in < or > to make a true statement.

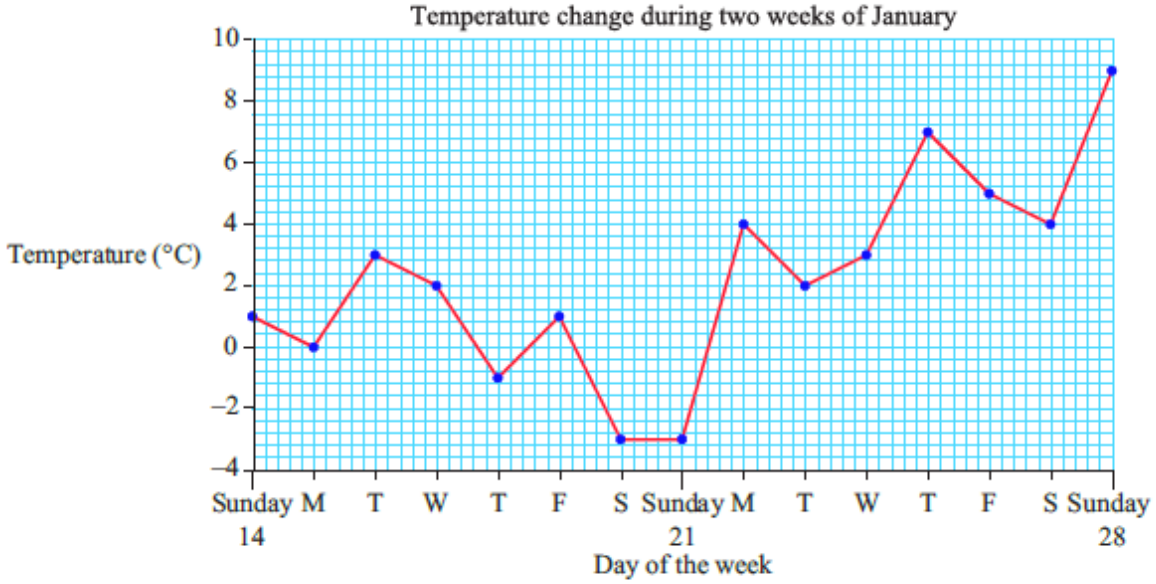
a 2□8	**b** 4□9	**c** 12□3	**d** 6□−4	**e** −7□4
f −2□4	**g** −2□−11	**h** −12□−20	**i** −8□0	**j** −2□2
k −12□−4	**l** −32□−3	**m** 0□−3	**n** −3□11	**o** 12□−89

2 Arrange each set of numbers in ascending order.

a −8, 7, 10, −1, −12 **b** 4, −3, −4, −10, 9, −8
c −11, −5, −7, 7, 0, −12 **d** −94, −50, −83, −90, 0

Applying your skills

3 Study the temperature graph carefully.



a What was the temperature on Sunday 14 January?
b By how much did the temperature drop from Sunday 14 to Monday 15?
c What was the lowest temperature recorded?
d What is the difference between the highest and lowest temperatures?
e On Monday 29 January the temperature changed by −12 degrees. What was the temperature on that day?

5 Mr Singh's bank account is $420 overdrawn.

a Express this as a directed number.
b How much money will he need to deposit to get his account to have a balance of $500?
c He deposits $200. What will his new balance be?

6 A diver 27 m below the surface of the water rises 16 m. At what depth is she then?

7 On a cold day in New York, the temperature at 6 a.m. was −5 °C. By noon, the temperature had risen to 8 °C. By 7 p.m. the temperature had dropped by 11 °C from its value at noon. What was the temperature at 7 p.m.?

8 Local time in Abu Dhabi is four hours ahead of Greenwich Mean Time. Local time in Rio de Janeiro is three hours behind Greenwich Mean Time.

a If it is 4 p.m. at Greenwich, what time is it in Abu Dhabi?
b If it is 3 a.m. in Greenwich, what time is it in Rio de Janiero?
c If it is 3 p.m. in Rio de Janeiro, what time is it in Abu Dhabi?
d If it is 8 a.m. in Abu Dhabi, what time is it in Rio de Janeiro?

EXPONENTS (POWERS) AND ROOTS

Whatch the following videos:
Introduction to exponents: http://shorturl.at/ansT4
Roots: http://shorturl.at/mrtGL

Powers and roots

Power and roots are terms that express a repeated multiplication of a number.

- **Powers**

E.g. $2 \times 2 \times 2 \times 2 \times 2 = 32$

2^5

We say this as 2 to the power of 5 because the number 2 is being multiplied by itself 5 times.

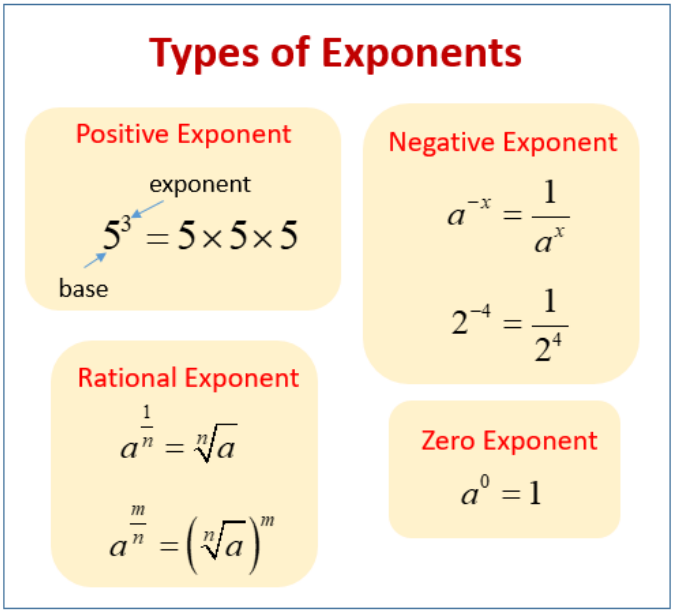
- **Roots**

E.g. $\sqrt[5]{32} = 2$.

As $2 \times 2 \times 2 \times 2 \times 2 \times = 32$, we can say that the 5th **root of** 32 is 2.

THIRD SPACE LEARNING

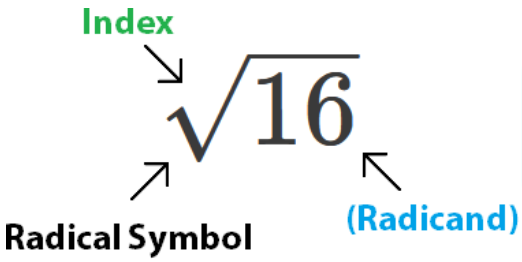
EXPONENTS



The power of 1	The power of 0
$a^1 = a$	$a^0 = 1$

Multiplying	To multiply two powers with the same base, add the indices.	$x^a \times x^b = x^{a+b}$
Dividing	To divide two powers with the same base, subtract the indices.	$x^a \div x^b = x^{a-b}$
Brackets	To raise one power to another power, multiply the indices.	$(x^a)^b = x^{ab}$
Fractions	To raise a base to a fraction $\frac{m}{n}$, calculate the nth root of the base, raised to the power m.	$x^{\frac{m}{n}} = \sqrt[n]{x^m}$
Negatives	To raise a base to a negative number, take the positive power of the reciprocal.	$x^{-a} = \frac{1}{x^a}$

ROOTS



$z^{\frac{5}{3}} = \sqrt[3]{z^5}$

"Fractional Exponents & Radicals"

ACTIVITY (LAST PAGES)

FRACTIONS

REVIEW:

Equivalent fractions

Find equivalent fractions by multiplying or dividing the numerator and denominator by the same number.

$\frac{1}{2} \times \frac{4}{4} = \frac{4}{8}$ $\frac{1}{2}$ and $\frac{4}{8}$ are equivalent

$\frac{40}{50} \div \frac{10}{10} = \frac{4}{5}$ $\frac{40}{50}$ and $\frac{4}{5}$ are equivalent

To simplify a fraction you divide the numerator and denominator by the same number.

$\frac{18}{40} = \frac{18 \div 2}{40 \div 2} = \frac{9}{20}$

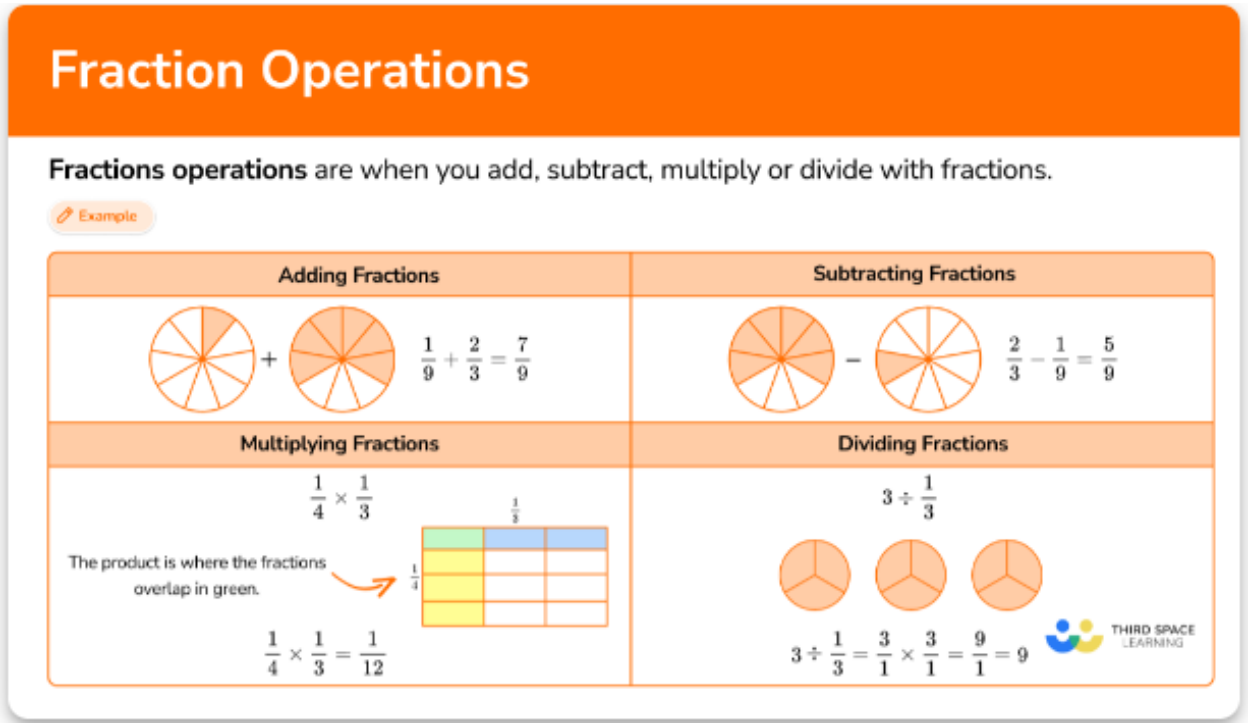
1 By multiplying or dividing both the numerator and denominator by the same number, find three equivalent fractions for each of the following.

a $\frac{5}{9}$ **b** $\frac{3}{7}$ **c** $\frac{12}{18}$ **d** $\frac{18}{36}$ **e** $\frac{110}{128}$

2 Express each of the following fractions in its simplest form.

a $\frac{7}{21}$ **b** $\frac{3}{9}$ **c** $\frac{9}{12}$ **d** $\frac{15}{25}$ **e** $\frac{500}{2500}$ **f** $\frac{24}{36}$ **g** $\frac{108}{360}$

Operations with fractions:



Evaluate each of the following.

1 a $\frac{2}{3}\times\frac{5}{9}$ b $\frac{1}{2}\times\frac{3}{7}$ c $\frac{1}{4}\times\frac{8}{9}$ d $\frac{2}{7}\times\frac{14}{16}$

2 a $\frac{50}{128}\times\frac{256}{500}$ b $1\frac{1}{3}\times\frac{2}{7}$ c $2\frac{2}{7}\times\frac{7}{8}$ d $\frac{4}{5}$ of $3\frac{2}{7}$

e $1\frac{1}{3}$ of 24 f $5\frac{1}{2}\times7\frac{1}{4}$ g $8\frac{8}{9}\times20\frac{1}{4}$ h $7\frac{2}{3}\times10\frac{1}{2}$

Evaluate the following.

1 a $\frac{1}{3}+\frac{1}{3}$ b $\frac{3}{7}+\frac{2}{7}$ c $\frac{5}{8}-\frac{3}{8}$ d $\frac{5}{9}+\frac{8}{9}$

e $\frac{1}{6}+\frac{1}{5}$ f $\frac{2}{3}-\frac{5}{8}$ g $2\frac{5}{8}-1\frac{3}{4}$ h $5\frac{1}{8}-3\frac{1}{16}$

2 a $4-\frac{2}{3}$ b $6+\frac{5}{11}$ c $11+7\frac{1}{4}$ d $11-7\frac{1}{4}$

e $3\frac{1}{2}-4\frac{1}{3}$ f $5\frac{1}{4}+3\frac{1}{16}+4\frac{3}{8}$ g $5\frac{1}{8}-3\frac{1}{16}+4\frac{3}{4}$ h $1\frac{1}{3}+2\frac{2}{5}-1\frac{1}{4}$

i $\frac{3}{7}+\frac{2}{3}\times\frac{14}{8}$ j $3\frac{1}{2}-2\frac{1}{4}\times\frac{4}{3}$ k $3\frac{1}{6}-1\frac{1}{2}+7\frac{3}{4}$ l $2\frac{1}{4}-3\frac{1}{3}+4\frac{1}{5}$

Evaluate each of the following.

1 $\frac{1}{7} \div \frac{1}{3}$

2 $\frac{2}{5} \div \frac{3}{7}$

3 $\frac{4}{9} \div 7$

4 $\frac{10}{11} \div 5$

5 $4\frac{1}{5} \div \frac{1}{7}$

6 $3\frac{1}{5} \div 5\frac{2}{3}$

7 $7\frac{7}{8} \div 5\frac{1}{12}$

8 $3\frac{1}{4} \div 3\frac{1}{2}$

1 $\frac{3}{4}$ of the people at an auction bought an item. If there are 120 people at the auction, how many bought something?

2 An essay contains 420 sentences. 80 of these sentences contain typing errors. What fraction (given in its simplest form) of the sentences contain errors?

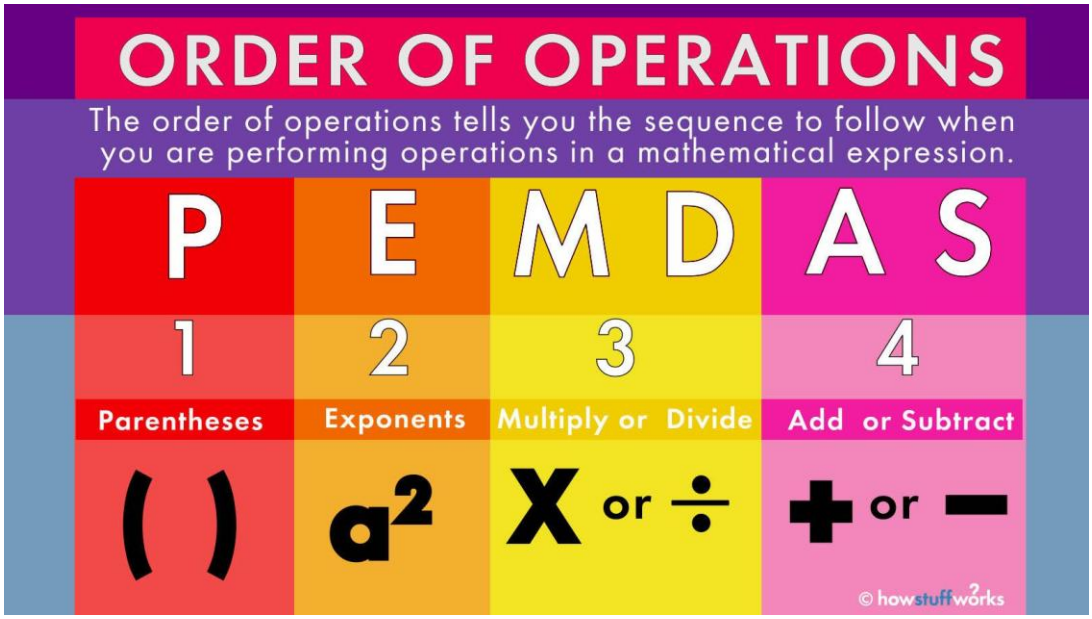
3 28 is $\frac{2}{7}$ of which number?

4 If $\frac{3}{5}$ of the people in a theatre buy a snack during the interval, and of those who buy a snack $\frac{5}{7}$ buy ice cream, what fraction of the people in the theatre buy ice cream?

5 Andrew, Bashir and Candy are trying to save money for a birthday party. If Andrew saves $\frac{1}{4}$ of the total needed, Bashir saves $\frac{2}{5}$ and Candy saves $\frac{1}{10}$, what fraction of the cost of the party is left to pay?

6 Joseph needs $6\frac{1}{2}$ cups of cooked rice for a recipe of Nasi Goreng. If 2 cups of uncooked rice with $2\frac{1}{2}$ cups of water make $4\frac{1}{3}$ cups of cooked rice, how many cups of uncooked rice does Joseph need for his recipe ? How much water should he add ?

ORDER OF OPERATIONS

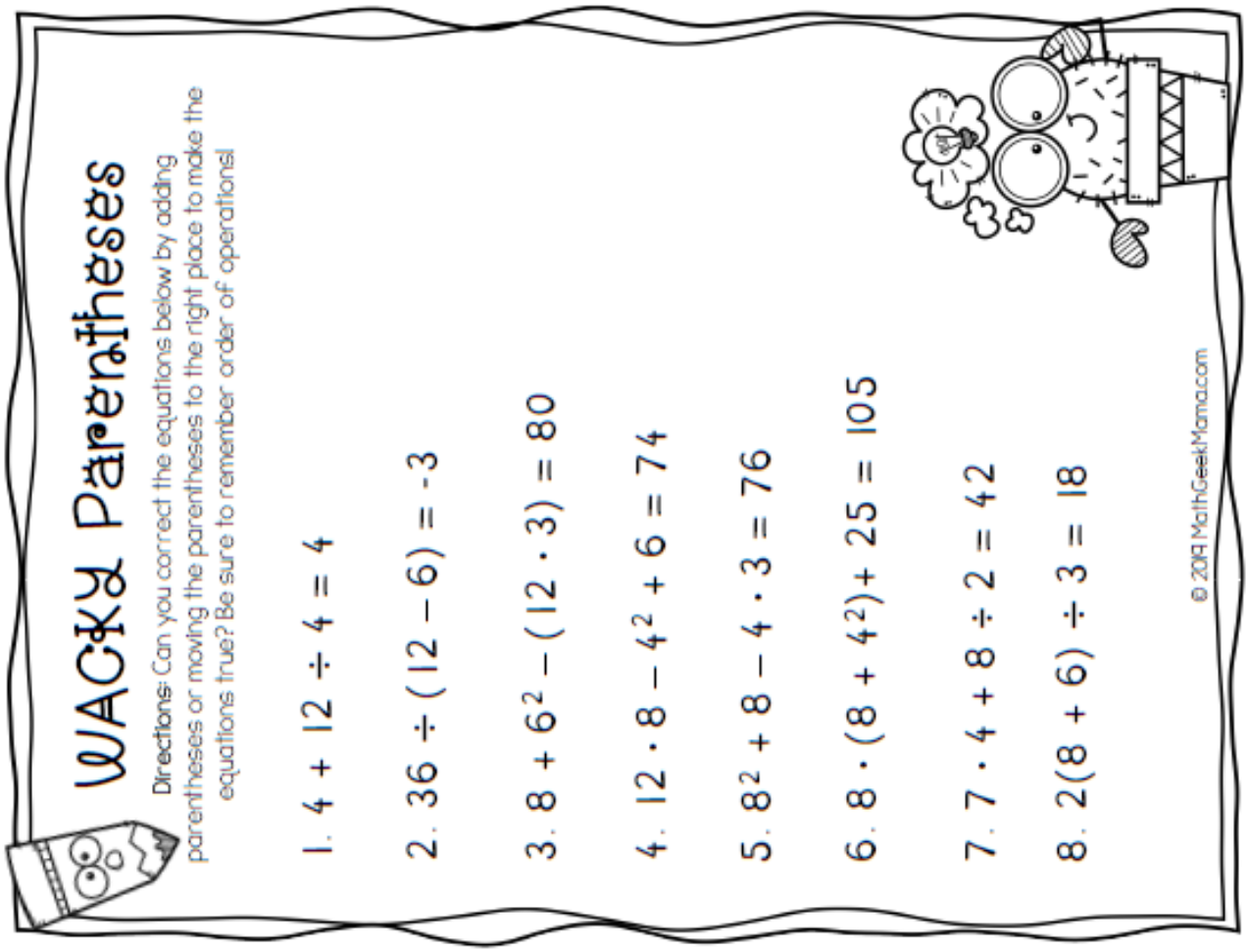


EXERCISES:

a) $5 - 2 \times 4 =$
b) $6 \times 3 + 12 \div 4 =$
c) $8 - 2 \times 3^2 =$
d) $7 + 9 \div 3 - 2 \times 4 =$
e) $2^2 + 3^3 \div 9 - 1 =$
f) $4 \times 3 + 8 \div 2 - 5 =$
g) $7 - 5 \times 2^2 + 6 =$
h) $3^3 - 2^2 \times 5 + 7 =$
i) $12 \div 4 - 1 \times 8 + 6 =$
j) $2^2 \times 3 + 5 - 4^3 \div 8$

WORD PROBLEMS

a) Maria has 15 toy cars. She wants to give some of them to her friends. If she gives 3 toy cars to each friend, how many friends can she give them to?
b) A delivery truck is carrying 240 boxes of cereal. If each box weighs 3 pounds, how much does the truck weigh in total?
c) A factory produced 125 widgets in 5 hours. If they continue to produce widgets at the same rate, how many widgets will they produce in 9 hours?
d) A pizza restaurant offers a special deal where customers can get a large pizza and a drink for $12. If a large pizza normally costs $10 and a drink normally costs $2, how much money will a customer save with the special deal?
e) A factory produces 500 items per day. If 20% of the items are defective, how many items are not defective?
f) Allison brought some t-shirts online. Each t-shirt cost $7. She had a $7 discount coupon. There was an additional charge of $4 per order for shipping costs. The total bill came to $60. How many t-shirts did Allison buy?
g)



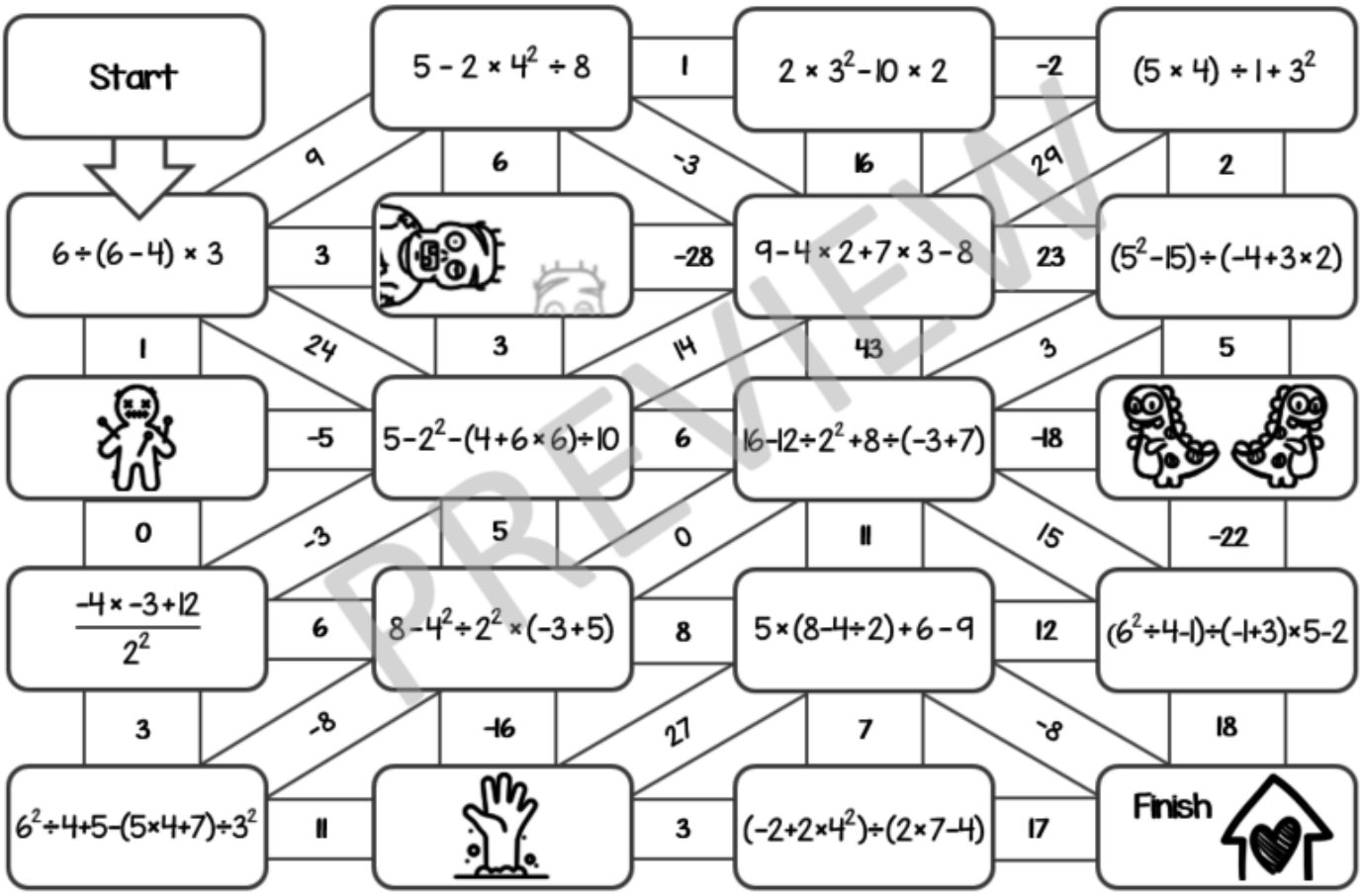
2+(8×3-6)+4×5-(28:2):2+16=	98 - 87 + (35-6) - (84:6) +9=
12x3+25-(8x9)-17+(47-32)=	37+15-43+(27:9)x(78-66)-32=
12+(50:2 +5x6 +3 -16)+7 -5=	4+(31-12)+(48-36)+45 -17 =
23+12-26x(45-42)-13+(8x4)=	(106-89)x(92:4-17)=
21x3+8-17+(85-12+9-45)=	(35-17):(8-6)+45=

Order of Operations Maze

Name: _____

Dodge the monsters. Make it home. Solve problems along the way to reveal the right path.
If a correct answer takes you to a monster, you made a wrong turn earlier on!

Created by ChalkDoc ↓ Creative Commons BY-NC 4.0 ↓ Icons by Freepik and Madebyoliver

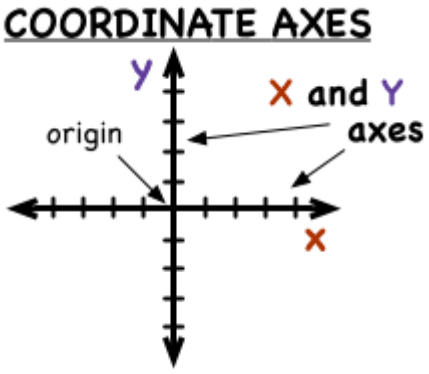


PREVIEW

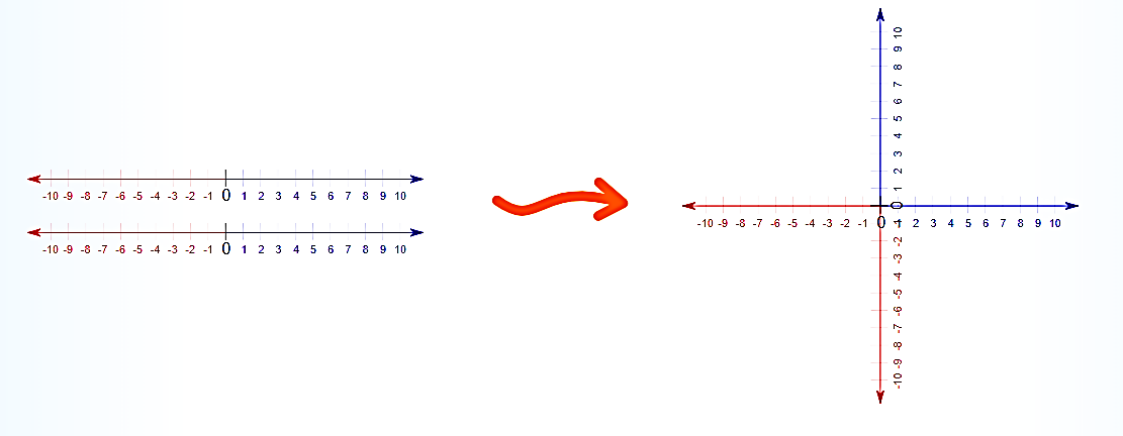
UNIT 2: ALGEBRA AND FUNCTIONS

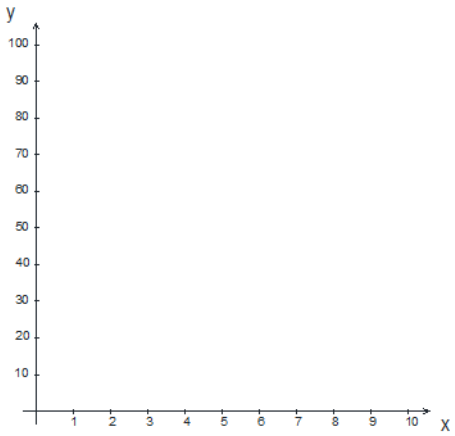
- **TWO-DIMENSIONAL COORDINATE SYSTEM**

A ***two-dimensional coordinate system*** is formed by two mutually perpendicular lines called *axes*. The axes intersect at the point O, which is called the ***origin***. In the right-handed system, one of the axes (x-axis) is directed to the right, the other y-axis is directed vertically upwards.



It is like we put two number lines put together, one going left-right, and the other going down-up. Remember that the tick marks in one number line must be equally apart from each other.





We can find axis which **scales** are different.

39

0. LOCATIONS:

The coordinates of any point on the xy-plane are determined by two real numbers x and y, which are orthogonal projections of the points on the respective axes. The x-coordinate of the point is called the abscissa of the point, and the y-coordinate is called its ordinate.

To describe the position or ***location*** of a point in relation to the origin you need coordinates (x,y). The first number tells you how far to move along in a horizontal direction. The second number tells you how far to move up or down in a vertical direction.

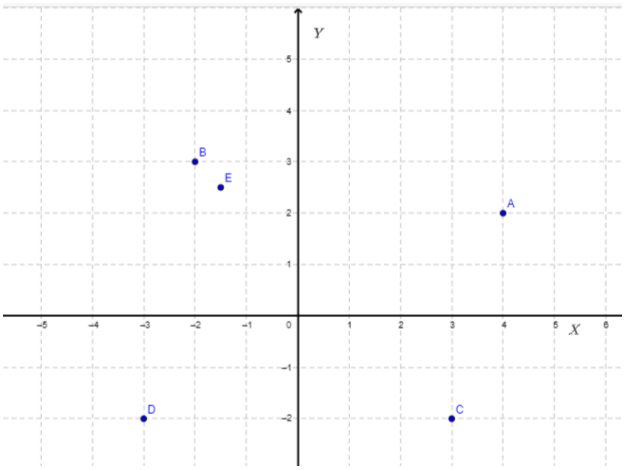
The origin has coordinates (0,0). It is where the axes intersect.

A positive *x-coordinate* means move right.

A positive *y-coordinate* means move up.

A negative *x-coordinate* means move left.

A negative *y-coordinate* means move down.



The coordinates of

A are (4,2)

B are (-2,3)

C are (3,-2)

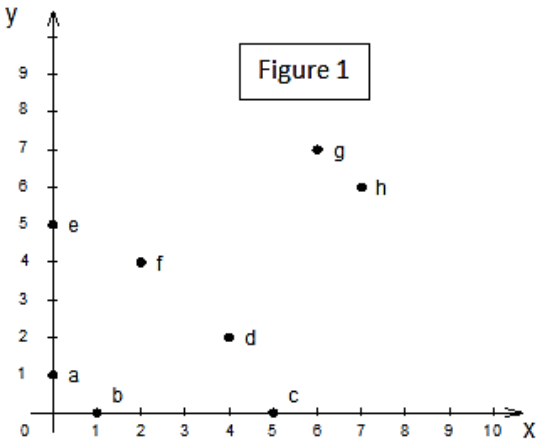
D are (-3,-2)

E are (1.5, 2.5)

EXERCISES

https://www.khanacademy.org/math/cc-sixth-grade-math/cc-6th-negative-number-topic/cc-6th-coordinate-plane/e/identifying_points_1

1. Draw a two-dimensional system in which you can plot the following locations with a proper scale: (0,5), (10,8), (50,6), (30,3).
2. Draw a two-dimensional system in which you can plot the following locations with a proper scale: (5,0), (8,10), (6,50), (3,30).
3. Draw a two-dimensional system in which you can plot the following locations with a proper scale: (500,0), (800,10), (600,50), (300,30).
4. Which of the following shows the coordinates for points "a" and "d" in *"Figure 1"*?



. (0,1) and (2,4)

. (0,1) and (4,2)

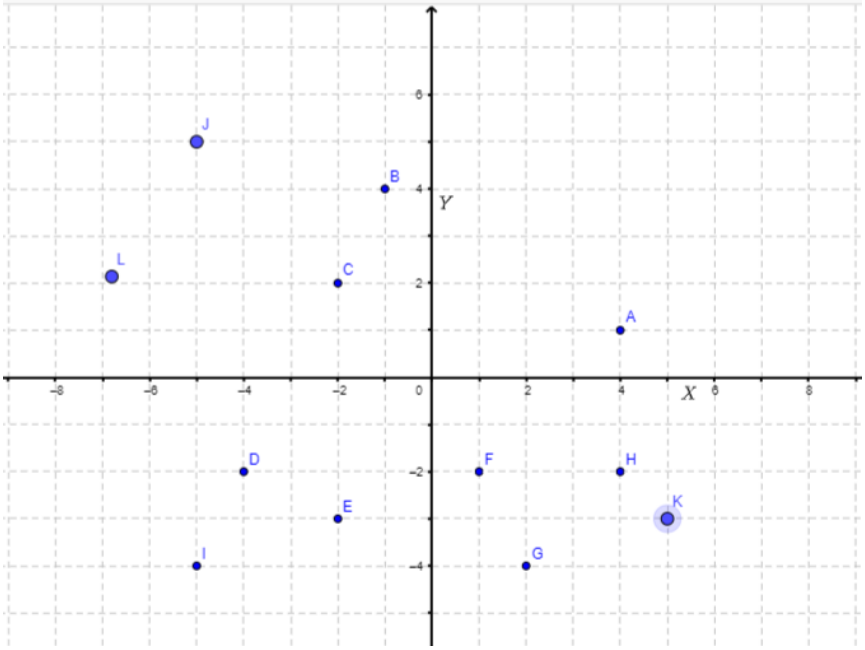
. (1,0) and (2,4)

. (1,0) and (4,2)

.

0. Which are the locations of points “b”, “c”, e”, “f”, “g” and “h” on “*Figure 1*”?

0. Write down the coordinates of the points “A” to “J” marked on the graph.



0. Draw a grid. Mark and label these points:

   A (3,2)

   B (-3,2)

   C (-3,-2)

   D (3,-2)

   Join the points A to B, B to C, C to D and D to A. What shape have you drawn?

0. Draw axes on graphed paper. On the x-axis, show the numbers: {-6,-5,…,5, 6}. On the y-axis, show the numbers: {-8,-7,…,7, 8}.

   Mark these points on the graph:

   (-3,-8), (-3,-7), (-5.-5), (-5,-4), (-4,4), (-4,-1), (-3,-1), (-3,5), (-2,5), (-2,-1), (-1,-1), (1,6), (2,6), (2,-3), (3, -3), (4,-1), (5,-1), (2,-7), (1,-7), (1,-8)

0. Draw a grid with *x*- and *y-axes* from -8 to 8.

. Plot the points A (-2,4), B (3,6), C (5,1).

. Plot the point D such that ABCD is a square. What are the coordinates of D?

. Draw the diagonals of the square. What are the coordinates of the point of intersection of the diagonals?

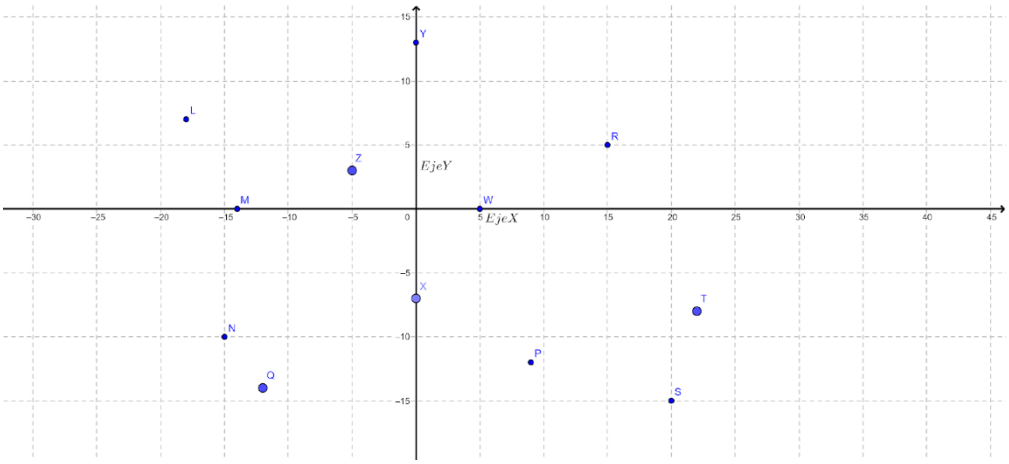
. Plot the points E (-4,-1), F (-6,-7), G (-2,-7).

. What is the shape EFG?

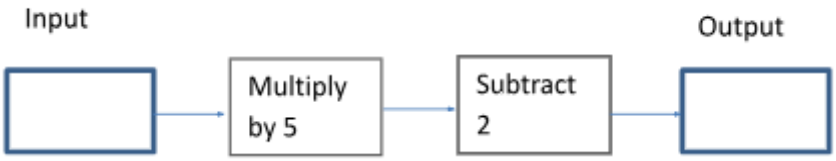
. Plot the point H at (-4,-7).

. Draw a line from E to H. What can you say about line EH?

0. Name the points with the following coordinates:



- Complete the table for the function machine:



Input	Output
1	
2	
3	
10	

-2	
-5	

- What are the numbers included in the domain? Write them as a set of numbers.

- What are the numbers included in the range? Write them as a set of numbers.

- Complete the mapping diagrams and express the domain and range as sets.

"Math Treasure Island":

Welcome to the exciting adventure of "Math Treasure Island"! In this project, you will explore Cartesian coordinates while searching for the hidden treasure. Follow these instructions to make the most of the game:

1. Preparation:

- Form groups of three: one pirate and two treasure seekers.
- The pirate hides the treasure on the map, marking it with the letter "T" and noting the coordinates. (Example: (3,-1)).

2. Map Creation:

- Each group receives grid paper to draw their own Cartesian plane.
- Label the X and Y axes on your sheet.

3. Treasure Hunt:

- In turns, each treasure seeker chooses a point on their own map.
- The pirate determines the color or letter based on the distance to the treasure:
    - **Green (G):** Within the first square around the treasure (e.g., (3, -2)).
    - **Orange (O):** Within the second square around the treasure.
    - **Red (R):** Within the third square around the treasure.
    - **Black (B):** Farther away than the three squares.
- Record the answers on your game sheets.

4. Recording and Analysis:

- After 7 turns, analyze the clues and try to deduce the coordinates of the treasure.
- Record all responses on your game sheets.

5. Declaration of Winners:

- If any treasure seeker discovers the treasure, they are the winner!
- If none find the treasure, the pirate reveals the correct coordinates.

6. Reflection and Evaluation:

- Reflect on how you applied coordinates during the game.
- Participate in a brief group discussion about what you learned.
- Record your experiences and conclusions on the project mural or presentation.

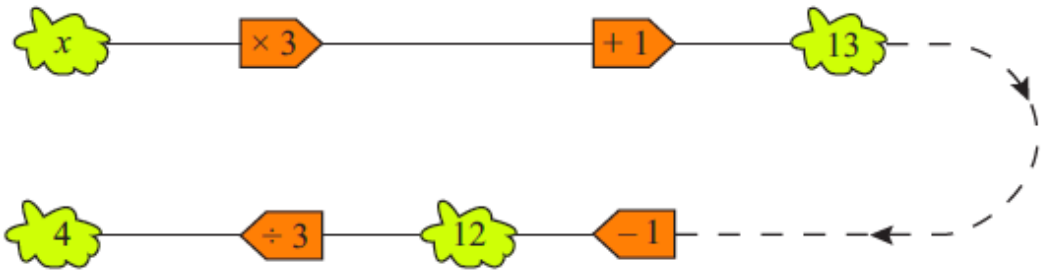
Enjoy "Math Treasure Island," and may the treasure hunt lead you to a full understanding of Cartesian coordinates!

LINEAR EQUATIONS

I think of a number. My number is x. If I multiply my number by three and then add one, the answer is 13. What is my number?

To solve this problem you first need to understand the stages of what is happening to x and then undo them in reverse order:

This diagram (sometimes called a function machine) shows what is happening to x, with the reverse process written underneath. Notice how the answer to the problem appears quite easily:



So $x = 4$

A more compact and efficient solution can be obtained using algebra. Follow the instructions in the question:

1	The number is x:	x
2	Multiply this number by three:	$3x$
3	Then add one:	$3x + 1$
4	The answer is 13:	$3x + 1 = 13$

This is called a **linear equation**. 'Linear' refers to the fact that there are no powers of x other than one.

The next point you must learn is that you can change this equation without changing the **solution** (the value of x for which the equation is true) *provided* you do the same to both sides at the same time.

Follow the reverse process shown in the function machine above but carry out the instruction on both sides of the equation:

$$3x + 1 = 13$$

$$3x + 1 - 1 = 13 - 1 \qquad \text{(Subtract one from both sides.)}$$

$$3x = 12$$

$$\frac{3x}{3} = \frac{12}{3} \qquad \text{(Divide both sides by three.)}$$

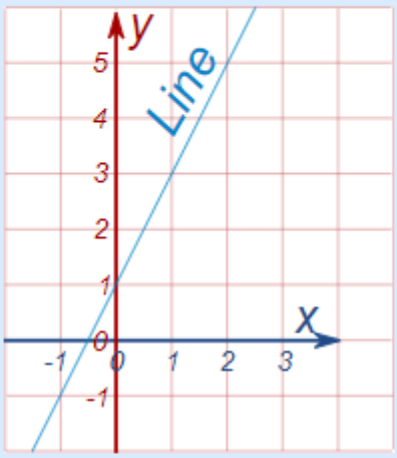
$$x = 4$$

Always line up your '=' signs because this makes your working much clearer.

Sometimes you will also find that linear equations contain brackets, and they can also contain unknown values (like x, though you can use any letter or symbol) on *both* sides.

A **linear** equation is an equation for a straight **line**

Example: **y = 2x + 1** is a linear equation:



The graph of **y = 2x+1** is a straight line

- When x increases, y increases **twice as fast**, so we need $2x$
- When x is 0, y is already 1. So $+1$ is also needed
- And so: $y = 2x + 1$

Here are some example values:

x	y = 2x + 1
-1	y = 2 × (-1) + 1 = **-1**
0	y = 2 × 0 + 1 = **1**
1	y = 2 × 1 + 1 = **3**
2	y = 2 × 2 + 1 = **5**

Check for yourself that those points are part of the line above!

But the variables (like "x" or "y") in Linear Equations do **NOT** have:

- Exponents (like the 2 in x^2)
- Square roots, cube roots, etc

Examples: These are **NOT** linear equations:

✗ $y^2 - 2 = 0$

✗ $3\sqrt{x} - y = 6$

✗ $x^3/2 = 16$

1 Solve the following equations.

a $4x + 3 = 31$ **b** $8x + 42 = 2$
c $6x - 1 = 53$ **d** $7x - 4 = -66$
e $9y + 7 = 52$ **f** $11n - 19 = 102$
g $12q - 7 = 14$ **h** $206t + 3 = 106$
i $\frac{2x+1}{3} = 8$ **j** $\frac{2x}{3} + 1 = 8$
k $\frac{3}{5}x + 11 = 21$ **l** $\frac{x+3}{2} = x$
m $\frac{2x-1}{3} = 3x$ **n** $\frac{3x}{2} + 5 = 2x$

2 Solve the following equations.

a $12x + 1 = 7x + 11$ **b** $6x + 1 = 7x + 11$ **c** $6y + 1 = 3y - 8$
d $11x + 1 = 12 - 4x$ **e** $8 - 8p = 9 - 9p$ **f** $\frac{1}{2}x - 7 = \frac{1}{4}x + 8$

3 Solve the following equations.

a $4(x + 1) = 12$ **b** $2(2p + 1) = 14$
c $8(3t + 2) = 40$ **d** $5(m - 2) = 15$
e $-5(n - 6) = -20$ **f** $2(p - 1) + 7(3p + 2) = 7(p - 4)$
g $2(p - 1) - 7(3p - 2) = 7(p - 4)$ **h** $3(2x + 5) - (3x + 2) = 10$

4 Solve for x.

a $7(x + 2) = 4(x + 5)$ **b** $4(x - 2) + 2(x + 5) = 14$
c $7x - (3x + 11) = 6 - (5 - 3x)$ **d** $-2(x + 2) = 4x + 9$
e $3(x + 1) = 2(x + 1) + 2x$ **f** $4 + 2(2 - x) = 3 - 2(5 - x)$

5 Solve the following equations for x

a $3^{3x} = 27$ **b** $2^{3x+4} = 32$
c $8.1^{4x+3} = 1$ **d** $5^{2(3x+1)} = 625$
e $4^{3x} = 2^{x+1}$ **f** $9^{3x+4} = 27^{4x+3}$

FIFA WORLD CUP 2026

Solve each problem by writing and solving a linear equation. Show all steps.

1) Ticket Revenue: A stadium sold 18,000 tickets. Some tickets cost $40 and others cost $65. If total revenue was $900,000, how many tickets of each type were sold?

2) Travel Time: A team travels at an average speed of 80 km/h. If the total distance was 560 km, how many hours did the trip take?

3) Group Stage Goals: A team scored 14 goals in total. If they scored 2 more goals in each of the last three matches than in the first match, how many goals did they score in the first match?

4) Match Duration: A match lasts 90 minutes plus extra time. If the total duration was 108 minutes, how many minutes of extra time were played?

5) Stadium Capacity: A stadium has 72,000 seats. If 8,000 seats are reserved and the rest are distributed equally in 8 sections, how many seats are in each section?

6) Team Points: A team earns 3 points for a win and 1 point for a draw. If a team earned 11 points after winning some matches and drawing 2, how many matches did they win?

7) Jersey Sales: A store sold World Cup jerseys for $55 each. If the total revenue was $4,400, how many jerseys were sold?

8) Training Sessions: A team trained for a total of 26 hours in one week. If they trained the same number of hours each day for 5 days, plus an extra 1 hour on the last day, how many hours did they train per regular day?

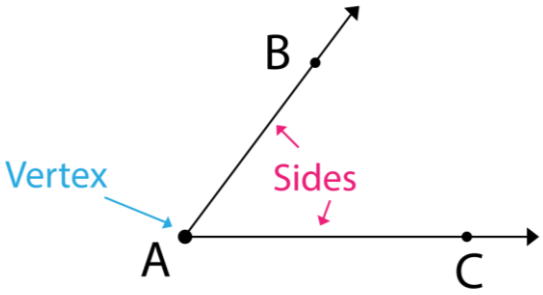
UNIT 3: GEOMETRY AND MEASUREMENTS

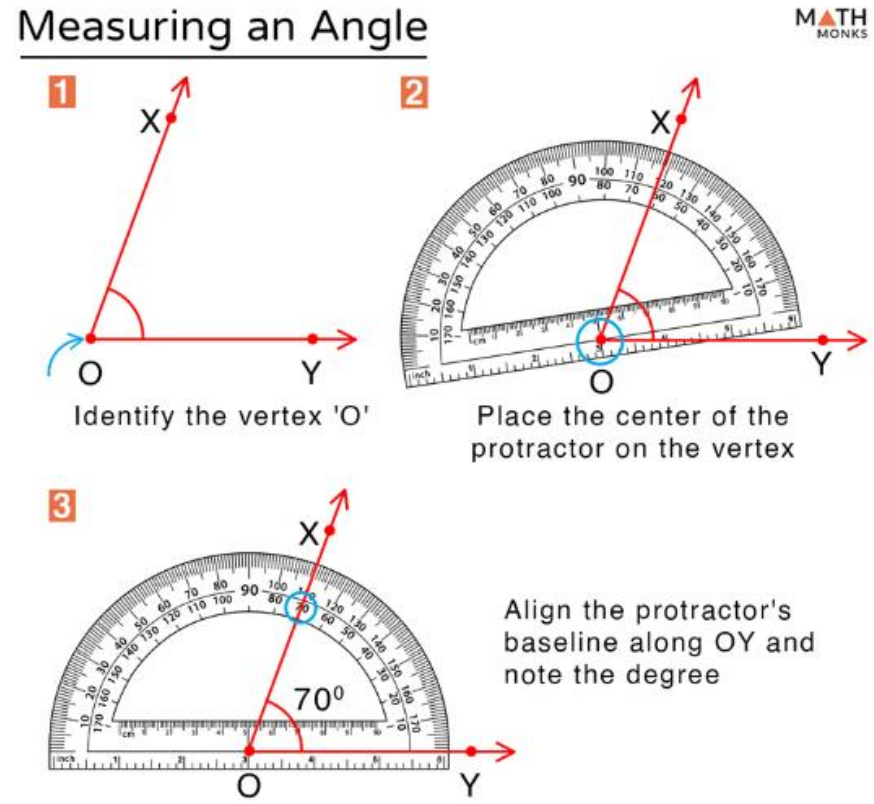
ANGLES IN A PLANE:

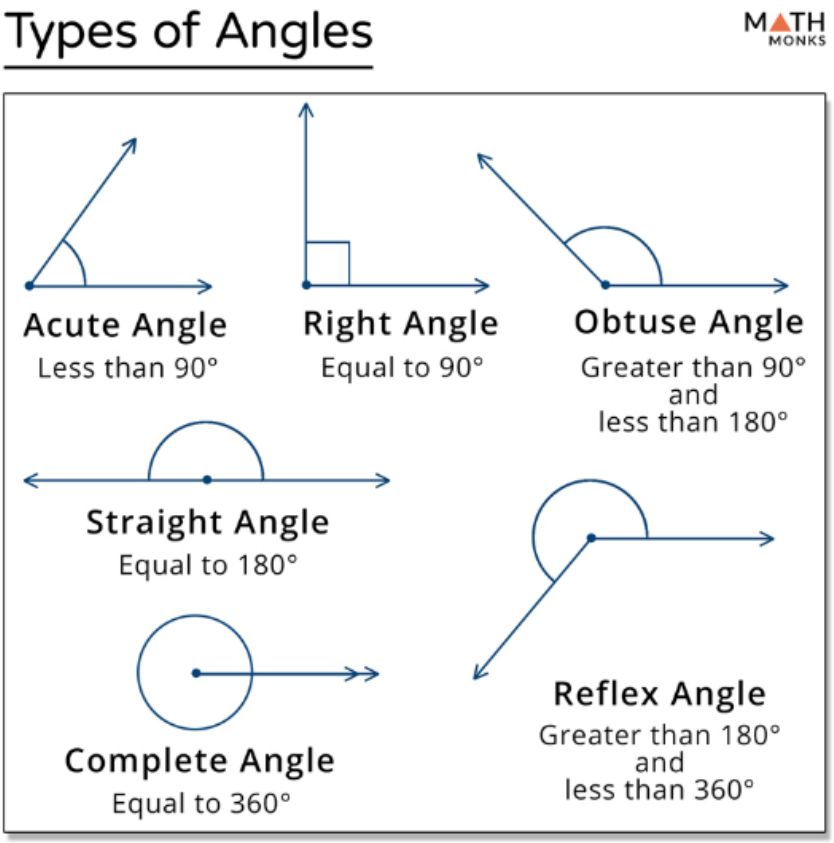
Mathematically, an angle is defined as a figure that forms when two rays meet at a common point. It is represented by the symbol ∠. An angle is usually measured in degrees, denoted with "°".

A degree is a measure of rotation. A full rotation around a point gives us a full circle which measures 360°, a half rotation gives us a half-circle, which is 180°, and a quarter rotation gives us a right-angle measuring 90°.

PARTS OF AN ANGLE



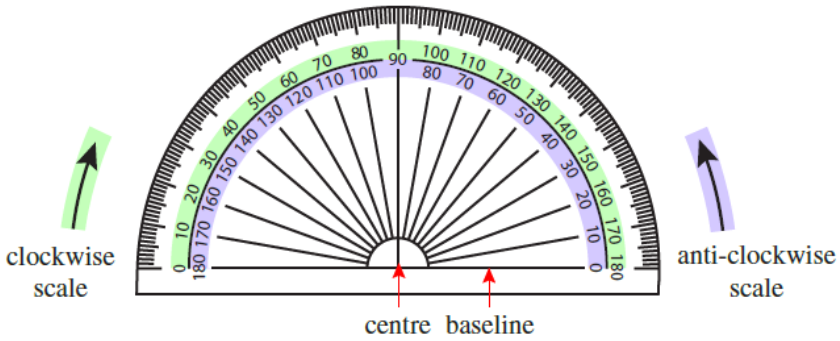

Measuring an Angle
MATH MONKS
1
X
O
Y
Identify the vertex 'O'
2
X
O
Y
Place the center of the protractor on the vertex
3
X
70⁰
O
Y
Align the protractor's baseline along OY and note the degree


Types of Angles
MATH MONKS
Acute Angle
Less than 90°
Right Angle
Equal to 90°
Obtuse Angle
Greater than 90° and less than 180°
Straight Angle
Equal to 180°
Complete Angle
Equal to 360°
Reflex Angle
Greater than 180° and less than 360°

ANGLES

Measuring and drawing angles

The size of an angle is the amount of turn from one arm of the angle to the other. Angle sizes are measured in degrees (°) from 0 to 360 using a protractor.



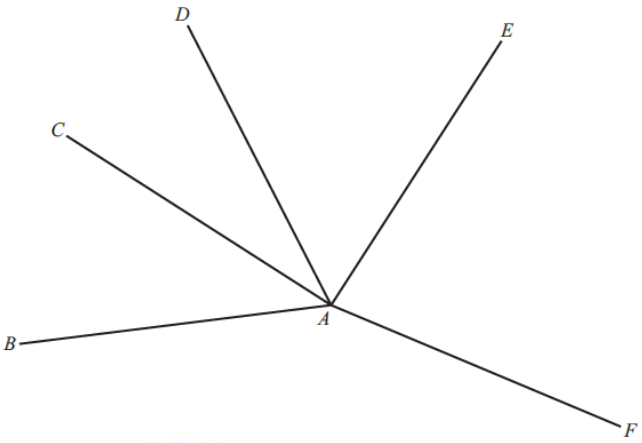
A 180° protractor has two scales. You need to choose the correct one when you measure an angle.

Measuring angles < 180°

Put the centre of the protractor on the vertex of the angle. Align the baseline so it lies on top of one arm of the angle.

Using the scale that starts with 0° to read off the size of the angle, move round the scale to the point where it crosses the other arm of the angle.

Exercise 3.1 **1** For each angle listed:



i *BAC*	**ii** *BAD*	**iii** *BAE*
iv *CAD*	**v** *CAF*	**vi** *CAE*
vii *DAB*	**viii** *DAE*	**ix** *DAF*

a state what type of angle it is (acute, right or obtuse)
b estimate its size in degrees
c use a protractor to measure the actual size of each angle to the nearest degree.
d What is the size of reflex angle *DAB*?

Exercise 3.2 Use a ruler and a protractor to accurately draw the following angles:

a $ABC = 80°$ **b** $PQR = 30°$ **c** $XYZ = 135°$

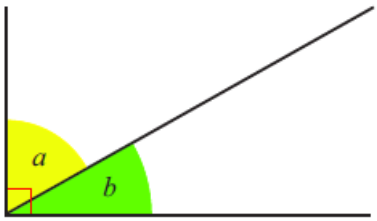
d $EFG = 90°$ **e** $KLM = 210°$ **f** $JKL = 355°$

Angle relationships

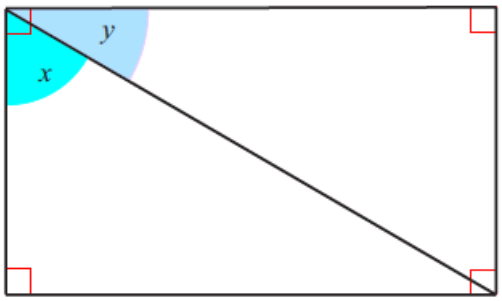
Make sure you know the following angle facts:

Complementary angles

Angles in a right angle add up to 90°.
When the sum of two angles is 90° those two angles are complementary angles.



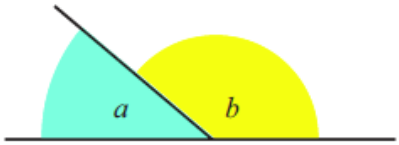
$a + b = 90°$



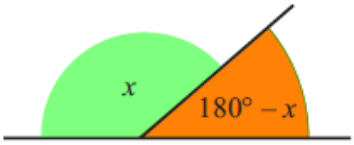
$x + y = 90°$

Supplementary angles

Angles on a straight line add up to 180°.
When the sum of two angles is 180° those two angles are supplementary angles.



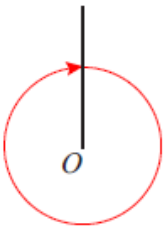
$a + b = 180°$



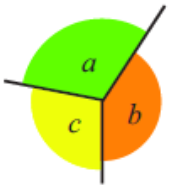
$x + (180° - x) = 180°$

Angles round a point

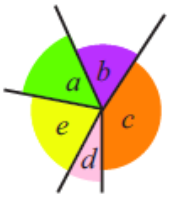
Angles at a point make a complete revolution.
The sum of the angles at a point is 360°.



360°



$a + b + c = 360°$

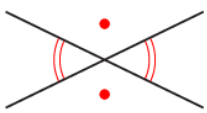


$a + b + c + d + e = 360°$

The adjacent angle pairs in vertically opposite angles form pairs of supplementary angles because they are also angles on a straight line.

Vertically opposite angles

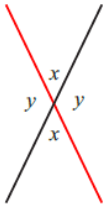
When two lines intersect, two pairs of **vertically opposite** angles are formed.
Vertically opposite angles are equal in size.



Two pairs of vertically opposite angles.

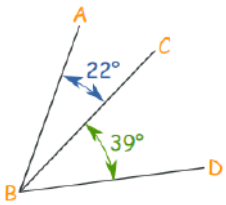


The angles marked x are equal to each other. The angles marked y are also equal to each other.



$x + y = 180°$

Adjacent Angles

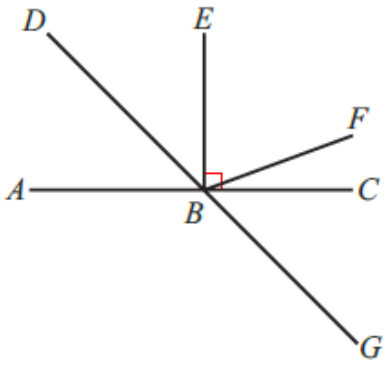


Two angles are **Adjacent** when they have a common side and a common vertex (corner point) and don't overlap.

Angle ABC is adjacent to angle CBD. Because: they have a common side (line CB) and they have a common vertex (point B).

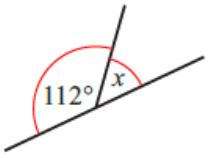
1 In the following diagram, name:

a a pair of complementary angles
b a pair of equal angles
c a pair of supplementary angles
d the angles on line *DG*
e the complement of angle *EBF*
f the supplement of angle *EBC*.

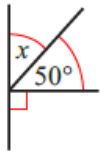


2 In each diagram, find the value of the angles marked with a letter.

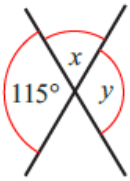
a



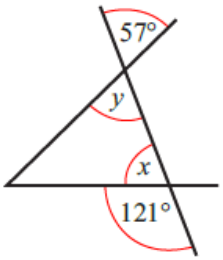
b



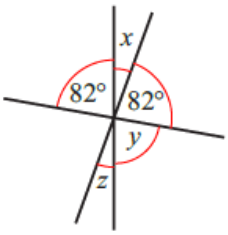
c



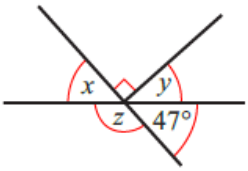
d



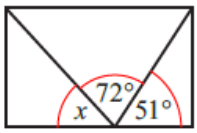
e



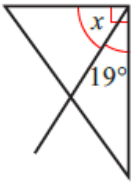
f



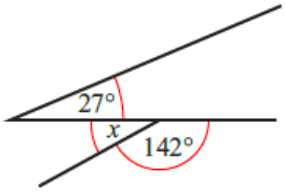
g



h

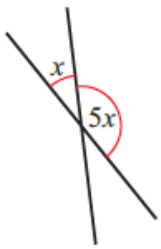


i

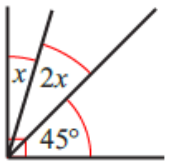


3 Find the value of x in each of the following figures.

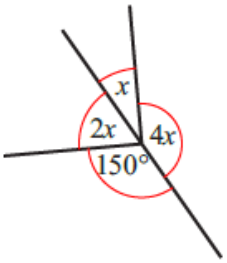
a



b



c



4 Two angles are supplementary. The first angle is twice the size of the second. What are their sizes?

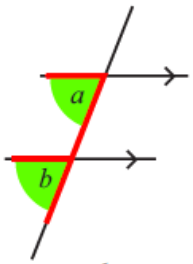
5 One of the angles formed when two lines intersect is 127°. What are the sizes of the other three angles?

Angles and parallel lines

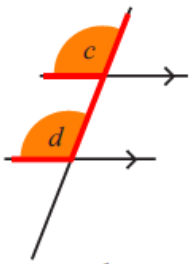
When two parallel lines are cut by a third line (the transversal) eight angles are formed. These angles form pairs which are related to each other in specific ways.

Corresponding angles ('F'-shape)

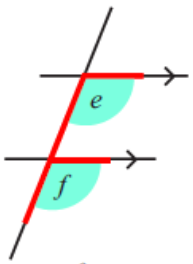
When two parallel lines are cut by a transversal four pairs of **corresponding** angles are formed. Corresponding angles are equal to each other.



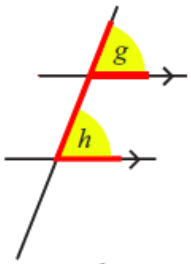
$a = b$



$c = d$



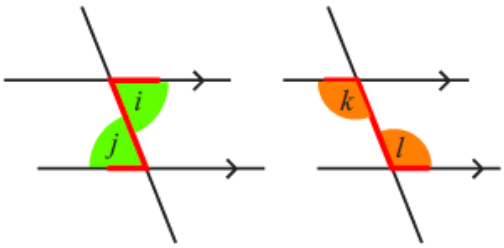
$e = f$



$g = h$

Alternate angles ('Z'-shape)

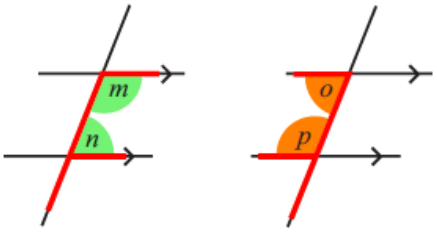
When two parallel lines are cut by a transversal two pairs of **alternate** angles are formed. Alternate angles are equal to each other.



$i = j$ $k = l$

Co-interior angles ('C'-shape)

When two parallel lines are cut by a transversal two pairs of **co-interior** angles are formed. Co-interior angles are supplementary (together they add up to 180°).

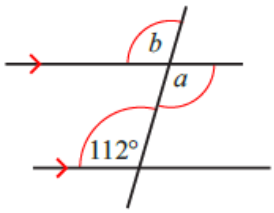


$m + n = 180°$ $o + p = 180°$

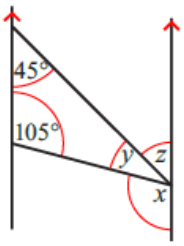
These angle relationships around parallel lines, combined with the other angle relationships from earlier in the chapter, are very useful for solving unknown angles in geometry.

1 Calculate the size of all angles marked with variables in the following diagrams. Give reasons.

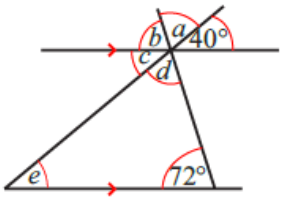
a



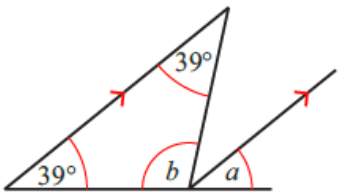
b



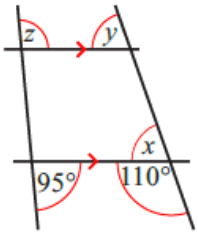
c



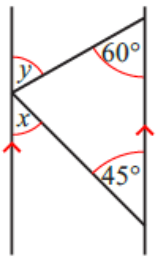
d



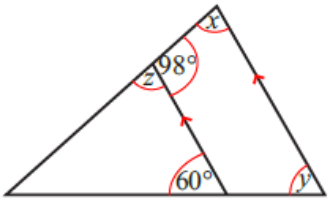
e



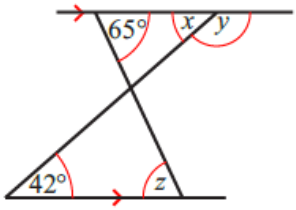
f



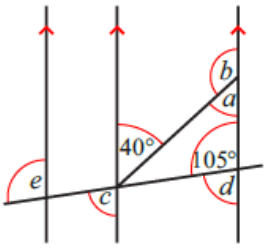
g



h

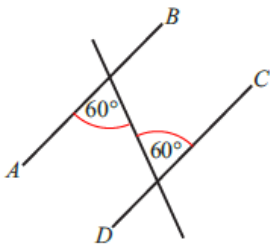


i

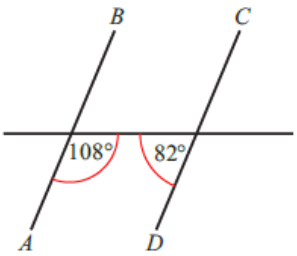


2 Decide whether $AB \| DC$ in each of these examples. Give a reason for your answer.

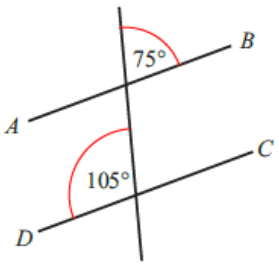
a



b



c



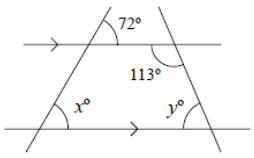
ANGLE

ANGLES WITHIN PARALLEL LINES

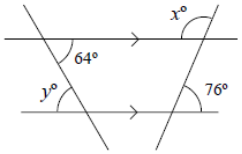
NO PROTRACTOR

Ref: G421. **3S1**

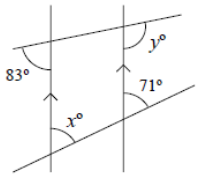
A1 Find the values of x and y



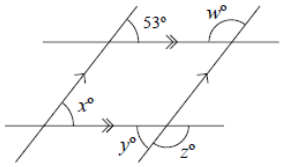
A2 Find the values of x and y



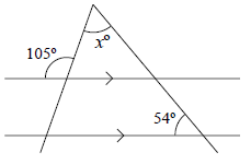
A3 Find the values of x and y



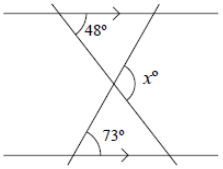
A4 Find the values of x, y, w and z



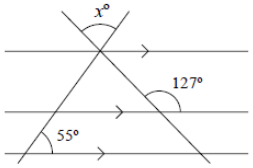
B1 Find the value of x



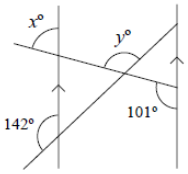
B2 Find the value of x



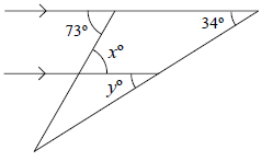
B3 Find the value of x



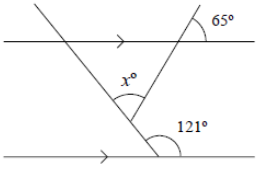
B4 Find the values of x and y



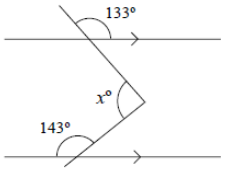
C1 Find the values of x and y



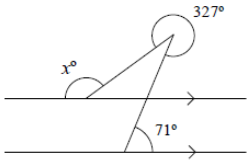
C2 Find the value of x



C3 Find the value of x



C4 Find the value of x



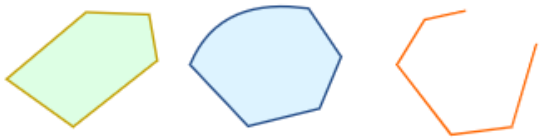
POLYGONS

A polygon is a plane shape with straight sides.

REVIEW: https://www.youtube.com/watch?v=IaoZhhx_I9s

Is it a Polygon?

Polygons are 2-dimensional shapes. They are made of straight lines, and the shape is "closed" (all the lines connect up).

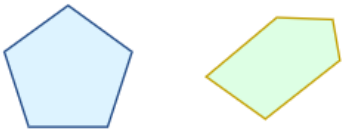


Polygon (straight sides) | **Not** a Polygon (has a curve) | **Not** a Polygon (open, not closed)

Types of Polygons

Regular or Irregular

A **regular** polygon has all angles equal and all sides equal, otherwise it is **irregular**

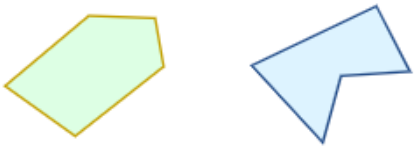


Regular | Irregular

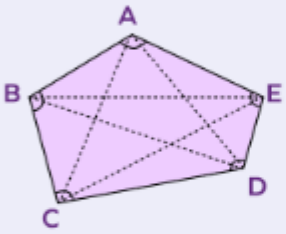
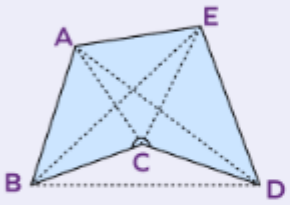
Concave or Convex

A **convex** polygon has no angles pointing inwards. More precisely, no internal angle can be more than 180°.

If any internal angle is greater than 180° then the polygon is **concave**. (*Think: concave has a "cave" in it*)

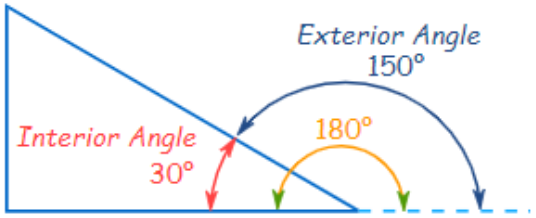


Convex Concave

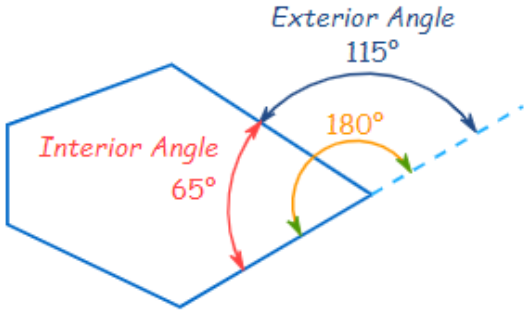
Convex Polygon	Concave Polygon
A convex polygon has all its interior angles less than 180 degrees.	A concave polygon has at least one interior angle more than 180 degrees.
A convex polygon has no diagonal in the exterior. 	Concave polygons have portions of diagonals in the exterior. 
All sides point outwards. There are no dents.	Some sides point inward. There are dents.
The line joining any two vertices of the convex polygon lies completely within it.	A line joining two vertices of a concave polygon may or may not lie within the polygon.
Convex polygons can be regular or irregular.	Concave polygons are never regular.

ANGLES IN A POLYGON

An Interior Angle is an angle inside a shape:



Another example:



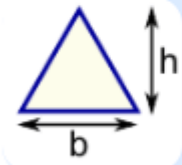
Sum of Interior Angles = (**n**−2) × 180°

Each Angle (of a Regular Polygon) = (**n**−2) × 180° / **n**

Area of Plane Shapes

Area is the size of a surface!

Learn more about Area, or try the Area Calculator.

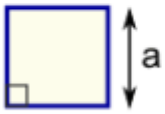


Triangle

Area = ½ × b × h

b = base

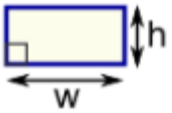
h = vertical height



Square

Area = a^2

a = length of side

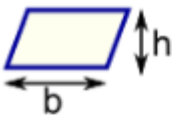


Rectangle

Area = w × h

w = width

h = height

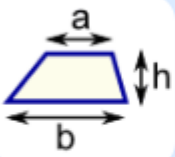


Parallelogram

Area = b × h

b = base

h = vertical height



Trapezoid (US)

Trapezium (UK)

Area = ½(a+b) × h

h = vertical height



Circle

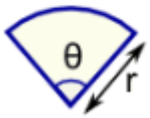
Area = **π** × r^2

r = radius



Ellipse

Area = **π**ab



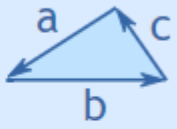
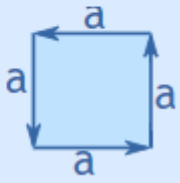
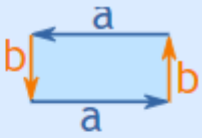
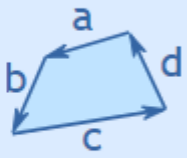
Sector

Area = ½ × r^2 × θ

r = radius

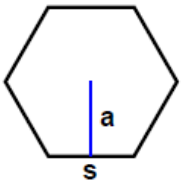
θ = angle in **radians**

Perimeter Formulas

	Triangle Perimeter = a + b + c
	Square Perimeter = 4 × a a = length of side
	Rectangle Perimeter = 2 × (a + b)
	Quadrilateral Perimeter = a + b + c + d
	Circle Circumference = 2πr r = radius

Identify and Calculate the Area and Perimeter for each Polygon.

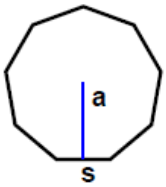
1)



s = 7.4 yds
a = 6.4086 yds

Area: _____
Perimeter: _____
Type: _____

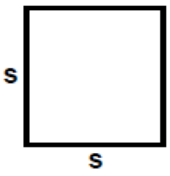
2)



s = 2.9 ft
a = 3.9838 ft

Area: _____
Perimeter: _____
Type: _____

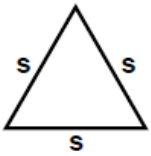
3)



s = 5.9 mm

Area: _____
Perimeter: _____
Type: _____

4)



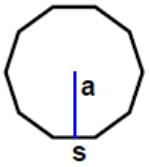
s = 6 inches

Area: _____

Perimeter: _____

Type: _____

5)



s = 2.6 mm

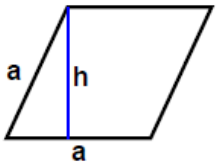
a = 4.001 mm

Area: _____

Perimeter: _____

Type: _____

6)



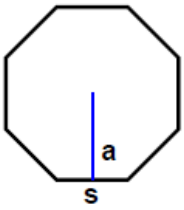
a = 6.2 inches h = 5.62 inches

Area: _____

Perimeter: _____

Type: _____

7)



s = 7.3 cm

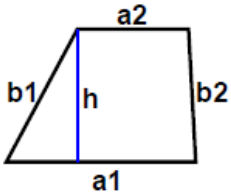
a = 8.8119 cm

Area: _____

Perimeter: _____

Type: _____

8)



a1 = 8 yds a2 = 4.7 yds

b1 = 6.35 yds b2 = 5.61 yds

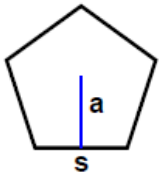
h = 5.6 yds

Area: _____

Perimeter: _____

Type: _____

9)



s = 6.3 cm

a = 4.3356 cm

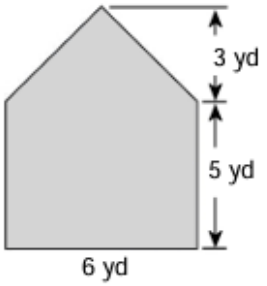
Area: _____

Perimeter: _____

Type: _____

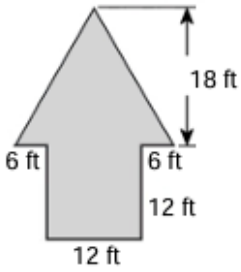
Find the area of each figure, round your answer to the nearest whole number if necessary.

1)



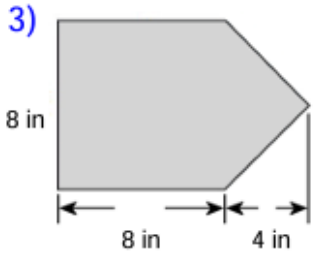
Area: _____

2)



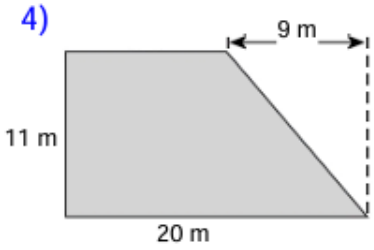
Area: _____

3)



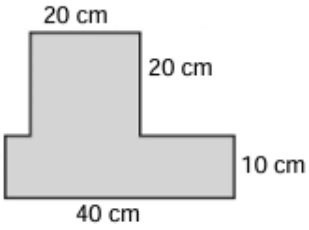
Area: _____

4)



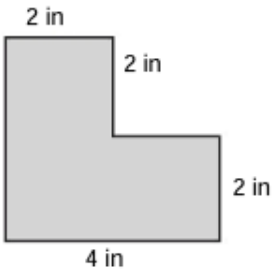
Area: _____

5)



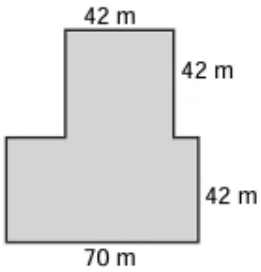
Area: _____

6)



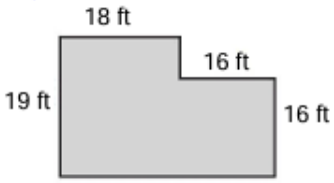
Area: _____

7)



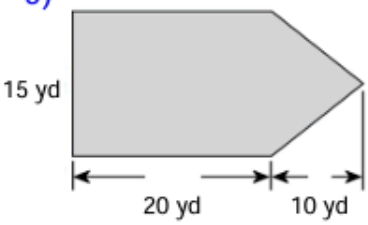
Area: _____

8)



Area: _____

9)



Area: _____

DATA

EXPLANATION VIDEO:

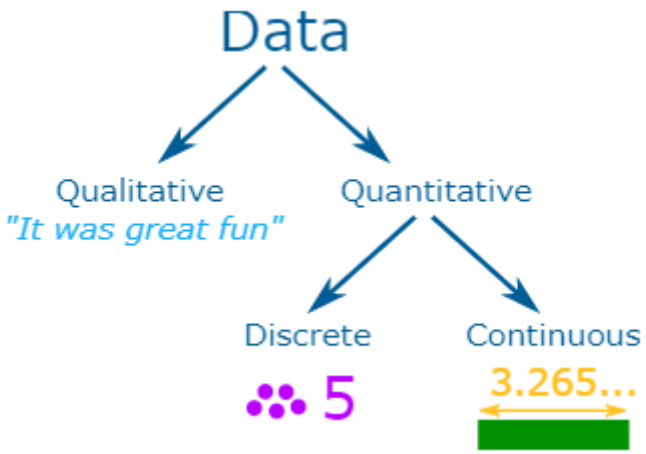
https://www.youtube.com/watch?v=hcgThf5mv38&ab_channel=mathantics

Data is a collection of facts, such as numbers, words, measurements, observations or just descriptions of things.

UNIT 4: DATA ANALYSIS, PROBABILITY AND STADISTICS

Data can be qualitative or quantitative.

- **Qualitative data** is descriptive information (it *describes* something)
- **Quantitative data** is numerical information (numbers)



Quantitative data can be Discrete or Continuous:

- **Discrete data** can only take certain values (like whole numbers)
- **Continuous data** can take any value (within a range)

Put simply: **Discrete data** is counted, **Continuous data** is measured.

MEAN, MEDIAN, AND MODE

Watch the following video:

http://shorturl.at/gkpF2

mean or average

Mean: Add up the numbers and divide by how many numbers.

Example: what is the central value of 3, 7 and 8?

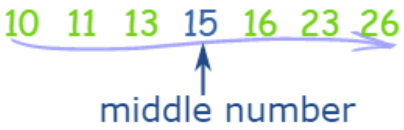
Answer: We calculate it by adding 3, 7 and 8 and then dividing the results by 3 (because there are 3 numbers):

(3+7+8) / 3 = 18/3 = 6



Median value

The Median is the ***"middle"*** of a sorted list of numbers.



How to Find the Median Value

To find the Median, place the numbers in **value order** and find the **middle**.

Example: find the Median of **12, 3 and 5**

Put them in order:

3, 5, 12

The middle is **5**, so the median is **5**.

Two Numbers in the Middle

BUT, with an **even amount of numbers** things are slightly different.

In that case we find the **middle pair** of numbers, and then find the value that is **half way** between them. This is easily done by adding them together and dividing by two.

Mode

The mode is simply the number which appears **most often**.

Example: 6, 3, 9, 6, 6, 5, 9, 3

In {6, 3, 9, 6, 6, 5, 9, 3} the Mode is 6, as it occurs most often.



More Than One Mode

We can have more than one mode.

Example: {1, 3, 3, 3, 4, 4, 6, 6, 6, 9}

3 appears three times, as does 6.

So there are two modes: at **3** and **6**

Having two modes is called **"bimodal"**.

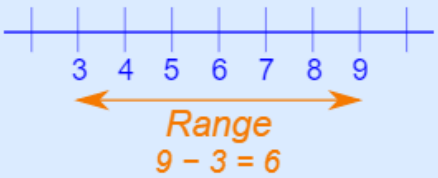
Having more than two modes is called **"multimodal"**.

RANGE

The Range is the difference between the lowest and highest values.

Example: In **{4, 6, 9, 3, 7}** the lowest value is 3, and the highest is 9.

So the range is 9 − 3 = **6**.



Find the mean, median, mode and range in each of the sets of data.
The first one has been done for you.

1)	15, 23, 19, 20, 23		5)	22, 37, 19, 25, 37, 51, 82	
order	15, 19, 20, 23, 23		order		
	Mean 100÷5=<u>20</u>	Median <u>20</u>		Mean	Median
	Mode <u>23</u>	Range 23-15=<u>8</u>		Mode	Range
2)	2, 7, 4, 2, 3, 6, 11		6)	6, 2, 13, 7, 6, 11, 10, 6, 2	
order			order		
	Mean	Median		Mean	Median
	Mode	Range		Mode	Range

3)	70, 63, 67, 62, 63		7)	109, 104, 96, 103, 104, 107, 98	
order			order		
	Mean	Median		Mean	Median
	Mode	Range		Mode	Range
4)	11, 4, 7, 8, 2, 6, 4		8)	14, 68, 38, 65, 36, 57, 65	
order			order		
	Mean	Median		Mean	Median
	Mode	Range		Mode	Range

STADISTICAL GRAPHS

Watch the following video:

http://shorturl.at/loWZ9

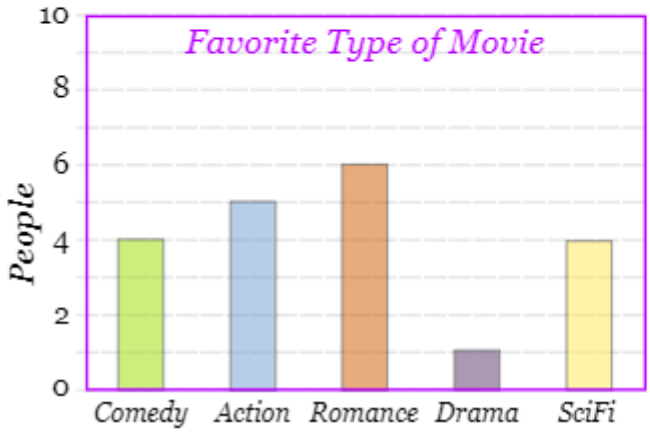
Who to show data?

A **Bar Graph** (also called Bar Chart) is a graphical display of data using bars of different heights.

Imagine you just did a survey of your friends to find which kind of movie they liked best:

Table: Favorite Type of Movie				
Comedy	**Action**	**Romance**	**Drama**	**SciFi**
4	5	6	1	4

We can show that on a bar graph like this:

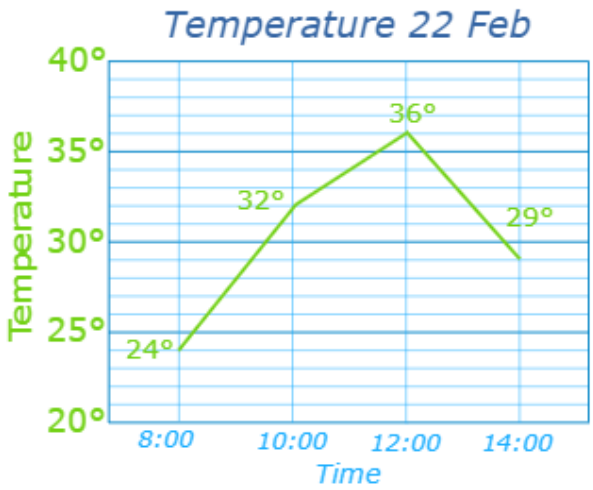


Line Graph: a graph that shows information connected in some way (usually as it changes over time).

You record the temperature outside your house and get these results:

Table: Temperature 22 Feb			
8:00	**10:00**	**12:00**	**14:00**
24°	32°	36°	29°

You are interested to see how it rises and falls, so decide to make a line graph:



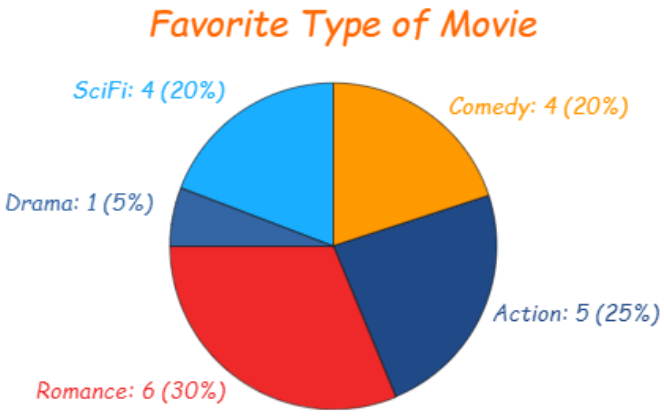
It makes the data come alive, right?
You can see how much it cooled down after lunch.

Pie Chart: a special chart that uses "pie slices" to show relative sizes of data.

Imagine you survey your friends to find the kind of movie they like best:

Table: Favorite Type of Movie				
Comedy	**Action**	**Romance**	**Drama**	**SciFi**
4	5	6	1	4

You can show the data by this Pie Chart:



It is a really good way to show relative sizes: it is easy to see which movie types are most liked, and which are least liked, at a glance.

How to Find the Mean

The mean is the **average** of the numbers.

It is easy to calculate: **add up** all the numbers, then **divide by how many** numbers there are

In other words it is the **sum** divided by the **count**.

Why Does This Work?

It is because 6, 11 and 7 added together is the same as 3 lots of 8:



Negative Numbers

How do you handle negative numbers? Adding a negative number is the same as subtracting the number (without the negative). For example 3 + (−2) = 3−2 = 1.

Knowing this, let us try an example:

Example 3: Find the mean of these numbers:

3, −7, 5, 13, −2

- The sum of these numbers is **3 − 7 + 5 + 13 − 2 = 12**
- There are **5** numbers.
- The mean is equal to **12 ÷ 5 = 2,4**

The mean of the above numbers is 2,4

Here is how to do it one line:

$$\text{Mean} = \frac{3 - 7 + 5 + 13 - 2}{5} = \frac{12}{5} = \mathbf{2,4}$$

The Median

But you could also use the Median: simply list all numbers in order and choose the middle one:

Example: Birthday Activities (continued)

List the ages in order:

1, 1, 1, 1, 1, 13, 13, 13, 13, 13, 13

Choose the middle number:

1, 1, 1, 1, 1, **13**, 13, 13, 13, 13, 13

The Median age is **13** ... so let's have a **Disco**!

Sometimes there are **two** middle numbers. Just average those two:

The Mode

The Mode is the value that occurs most often:

Example: Birthday Activities (continued)

Group the numbers so we can count them:

1, 1, 1, 1, 1, 13, 13, 13, 13, 13, 13

"13" occurs 6 times, "1" occurs only 5 times, so the mode is **13**.

How to remember? Think "mode is most"

But Mode can be tricky, there can sometimes be more than one Mode.

Frequency

Frequency is how often something occurs.

Example: Sam played football on:



- Saturday Morning,
- Saturday Afternoon
- Thursday Afternoon

The frequency was 2 on Saturday, 1 on Thursday and 3 for the whole week.

Frequency Distribution

By counting frequencies we can make a **Frequency Distribution** table.

Example: Goals

Sam's team has scored the following numbers of goals in recent games

2, 3, 1, 2, 1, 3, 2, 3, 4, 5, 4, 2, 2, 3

Sam put the numbers in order, then added up:

- how often 1 occurs (2 times),
- how often 2 occurs (5 times),
- etc,

and wrote them down as a **Frequency Distribution** table.

From the table we can see interesting things such as

- getting 2 goals happens most often
- only once did they get 5 goals

Scores:
1,1,2,2,2,2,2,3,3,3,3,4,4,5

Score	Frequency
1	2
2	5
3	4
4	2
5	1

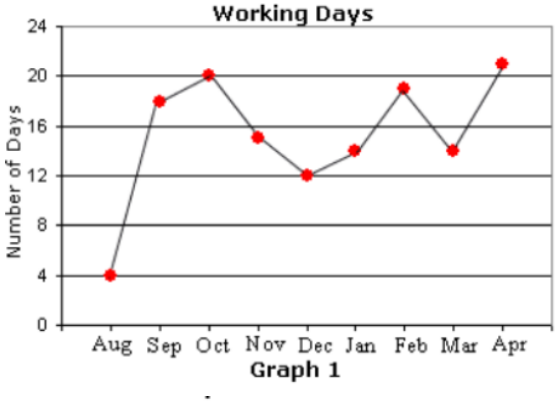
This is the definition:

Frequency Distribution: values and their frequency (how often each value occurs).

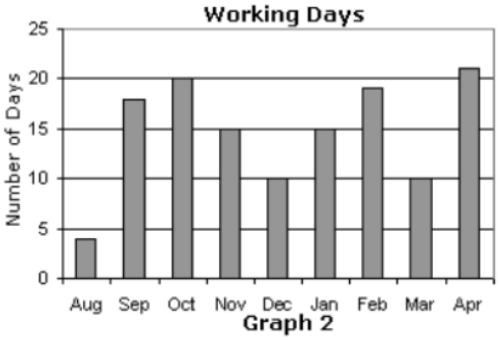
1. The table shows the number of school days each month during the last school year. Which graph best represents this data?

   a. Graph 1
   b. Graph 2
   c. Graph 3
   d. Graph 4

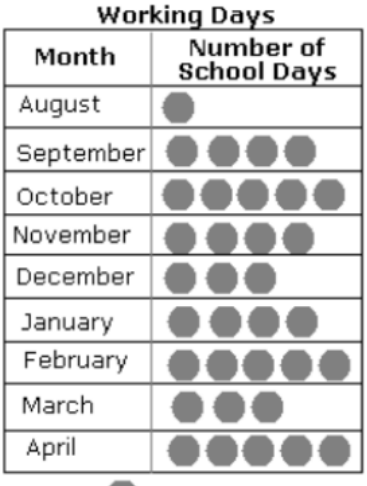
Month	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Number of Days	4	18	20	15	12	14	19	14	21



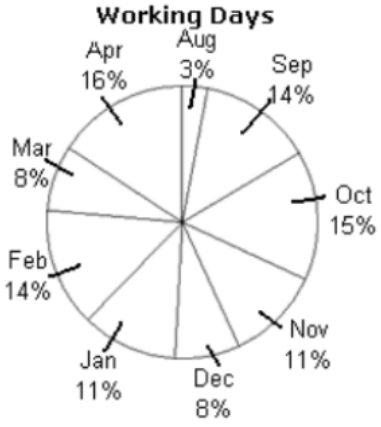
Graph 1



Graph 2



Graph 3



Graph 4

2. Identify the graph that is most appropriate to represent the amount of money spent on military over a period of time.
   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

3. Which graph would be most appropriate to show the time in number of minutes that 25 employees take to travel to the office?
   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

4. A survey was conducted on the length of the names of the students in a class. What type of graph would be most appropriate to show the range of the data?
   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

5. The table shows the kinds of pets the students have in a class. What type of graph would be most appropriate for the data?

Pet	Frequency of the Pet
Birds	18
Cats	5
Dogs	23
Fish	17
Rabbits	6
Horse	4

   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

6. The table shows the yearly average earnings (in thousands of dollars) for men and women. Which type of graph would be most appropriate for the data?

Year	1998	1999	2000	2001
Men	28.9	29.3	29.6	30.2
Women	19.5	20.4	22.3	22.4

a. Line graph
b. Bar graph
c. Pie graph
d. Histogram
e. Scatter plot

7. The weight of 15 people(in kg) are listed: 34, 49, 23, 54, 35, 64, 40, 53, 52, 76, 48, 21, 47, 34, and 33. Which type of graph is appropriate for the data?

a. Line graph
b. Bar graph
c. Pie graph
d. Histogram
e. Scatter plot

8. The table shows the number of cars sold in a week. Which type of graph is appropriate for the data?

a. Line graph
b. Bar graph
c. Pie graph
d. Histogram
e. Scatter plot

Color of car	Red	White	Orange	Blue
Number of cars	15	24	7	11

9. The table shows the school enrollment between 1992 to 1996. Which type of graph is appropriate for the data?

Year	Number of students
1992	65
1993	85
1994	70
1995	70
1996	75

a. Line graph
b. Bar graph
c. Pie graph
d. Histogram
e. Scatter plot

Project Guidelines for "Exploring stadistics with Lionel Messi":

Introduction:

- This activity aims to explore and apply the concepts of mode, median, and mean through the analysis of stadistics related to Lionel Messi's career.

Step 1: Topic Selection (10 minutes):

- Choose "The Career of Lionel Messi" as your main theme for stadistical research.
- Briefly explain why you chose this topic and how you think stadistics can reveal interesting aspects of Messi's career.

Step 2: Research (15 minutes):

- Use electronic devices to search and compile relevant stadistics about Lionel Messi.
- Make sure to gather at least 15 data points, which could include goals per season, assists, titles won, etc.
- Record the source of each stadistic for reference.

Step 3: Analysis and Calculations (10 minutes):

- Perform calculations to determine the mode, median, and mean of the collected data.
- You can work in small groups to discuss and compare results.

Step 4: Creation of Graphs (5 minutes):

- Use online tools to create graphs representing the selected data. Choose the type of graph that best highlights the chosen stadistics.
- Include the graphs in your report.

Step 5: Written Presentation (5 minutes):

- Prepare a written report that includes:
    - A brief introduction about Lionel Messi.
    - Presentation of the selected stadistics.
    - How you calculated the mode, median, and mean.
    - Include the created graphs.
    - Reflections on what you learned through this activity.

Submission and Evaluation:

- Submit your reports at the end of the class.
- Evaluation will be based on the quality of the written presentation, accuracy in calculations, and creativity in graph creation.

Important Note:

- The written presentation should be clear and organized. Make sure to explain each step and provide sufficient details for anyone to understand your findings.
- Enjoy exploring Lionel Messi's stadistics and discovering the fascinating world of stadistics applied to sports!

Probability

*How **likely** something is to happen.*

Many events can't be predicted with total certainty. The best we can say is how **likely** they are to happen, using the idea of probability.

Tossing a Coin



When a coin is tossed, there are two possible outcomes:

Heads (H) or
Tails (T)

Also:

- the probability of the coin landing **H** is ½
- the probability of the coin landing **T** is ½

Throwing Dice



When a single die is thrown, there are six possible outcomes: **1, 2, 3, 4, 5, 6**.

The probability of any one of them is $\frac{1}{6}$

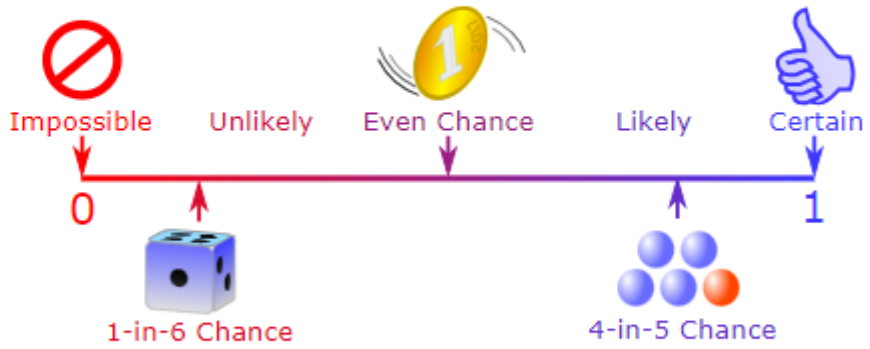
Probability

In general:

$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Probability Line

We can show probability on a Probability Line:



Probability is always between 0 and 1

Words

Some words have special meaning in Probability:

Experiment: a repeatable procedure with a set of possible results.

Example: Throwing dice



We can throw the dice again and again, so it is repeatable.

The set of possible results from any single throw is {1, 2, 3, 4, 5, 6}

Outcome: A possible result.

Example: "6" is one of the outcomes of a throw of a die.



Trial: A single performance of an experiment.

Example: I conducted a coin toss experiment. After 4 trials I got these results:

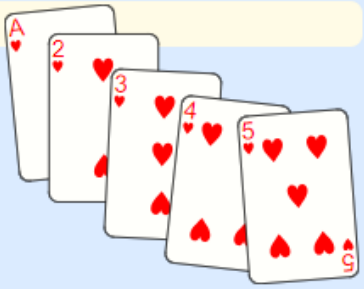


Outcome	Trial	Trial	Trial	Trial
Head	✓	✓		✓
Tail			✓	

Three trials had the outcome "Head", and one trial had the outcome "Tail"

Sample Space: all the possible outcomes of an experiment.

Example: choosing a card from a deck



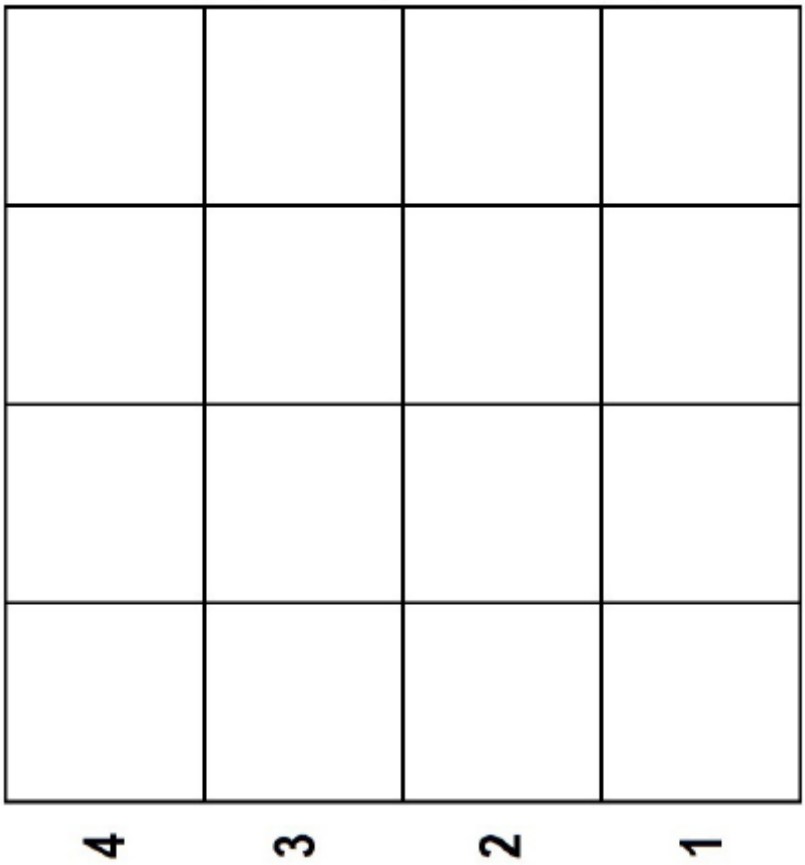
There are 52 cards in a deck (not including Jokers)

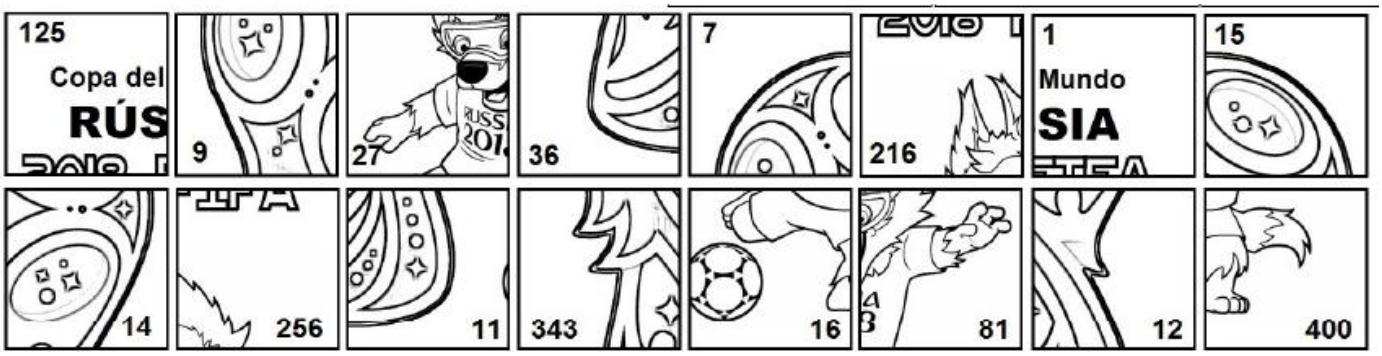
So the **Sample Space is all 52 possible cards**: {Ace of Hearts, 2 of Hearts, etc... }

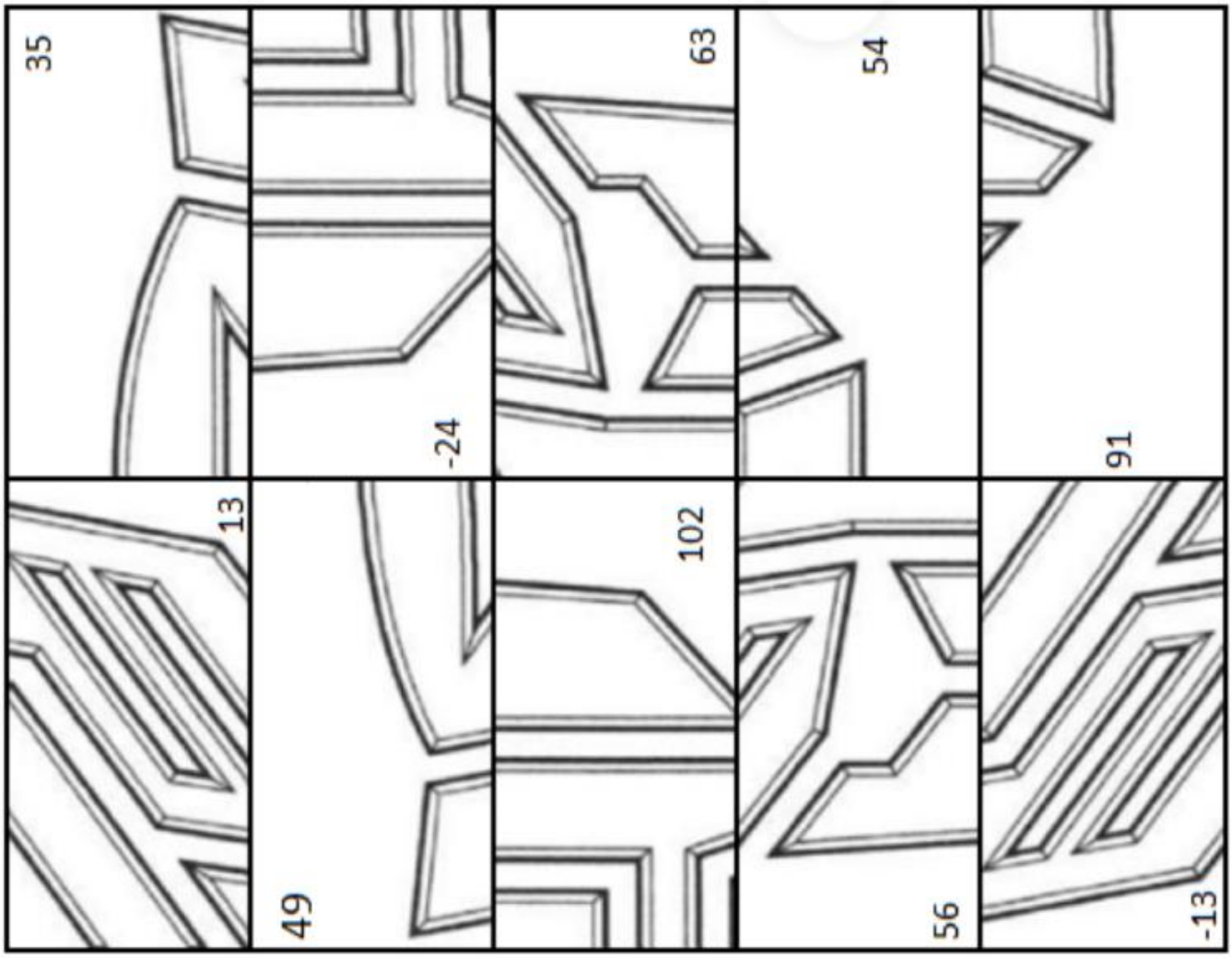
ACTIVITIES

10	1	11	7^2
11^2	1^2	8^2	$\sqrt{121}$
12	9	11	2
4^2	$\sqrt{1}$	12^2	3^2
121	25	9	144
$\sqrt{144}$	$\sqrt{16}$	$\sqrt{81}$	9^2
64	5	81	9
5^2	$\sqrt{49}$	$\sqrt{64}$	6^2
49	4	8	100
$\sqrt{9}$	2^2	6^2	$\sqrt{36}$
1	3	36	16
$\sqrt{100}$	10^2	$\sqrt{25}$	$\sqrt{4}$

(C;4) $5^9 : 5^6$	(B;3) $\sqrt{196}$	(D;1) $20^{10} : 20^8$					
(A;3) $\sqrt{81}$	(D;4) $(2416)^0$	(A;1) 6^2					
(D;2) $(3^2)2$	(A;2) 7^3	(B;2) $\sqrt{144}$					
(B;1) $\sqrt{121}$	(C;2) $3^2 \times 3^1$	(C;1) $\sqrt{256}$					
(B;4) $\sqrt{225}$	(D;3) $(2^2)4$	(A;4) $\sqrt{49}$					
		(C;3) $6^2 \times 6^1$					


D
C
B
A
4
3
2
1


125
Copa del
RÚS
9
27
36
7
216
1
Mundo
SIA
15
14
256
11
343
16
81
12
400


35
-24
63
54
91
13
49
102
56
-13

- Calculate the fractions and color the path of the solutions to guide "Santa's Little Helper" with Homer and Bart.



$2\frac{3}{4} + \frac{3}{4} =$

$2 - \frac{3}{4} =$

$3 + \frac{1}{10} =$

$4\frac{1}{3} - \frac{1}{6} =$

$5\frac{2}{4} - \frac{2}{2} =$

$3\frac{1}{4} + \frac{1}{8} - \frac{1}{2} =$